

IndusLink

Final Year Project Report

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The University of Faisalabad Degree
of
Bachelor of Science in **Software Engineering**



Department of Computer Science
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Project Details

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Declaration

We here by declare that the content of this project report title “**IndusLink**” submitted to the “**DEPARTMENT OF COMPUTER SCIENCE**”, is a documentation of a unique work we created under the supervision of Supervisor “**Dr. Usman Ahmed**” and that no part has been plagiarized (except the references, some standard mathematical or genetic models/questions/protocols, etc.). Additionally, this project is presented in partial completion of the degree requirements for a Bachelor of Science in “**Software Engineering**”. The University may take action if the above statement is found inaccurate at any stage.

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Abstract

In Pakistan, textile industry face significant collaborative challenges because there is no centralized digital platform for inter-factory collaboration and contract agreements. Textile industry has multiple departments such as dyeing, knitting, printing and stitching etc. Mostly factories do not have all production units in-house so, they have to coordinate with other factories to fulfill their orders. Moreover, to locate best service providers factory owners and managers visits other factories physically for finalization and contracts approval which is firstly time-consuming and secondly there is no standard method to store business dealings digitally. The primary purpose of IndusLink is to make collaboration digitally between sellers and manufacturers in textile sector. This web-based platform offers the features such as role-based interface, profile creation, create and approve contracts digitally, place orders and receive invoices, integrated real time chat system and find the best service provider. This web-based platform is developed using modern technologies such as Next.js for frontend, Python Fast API for backend and Supabase serves as the database to store overall data. This platform reduces physical visits, improve transparency in agreements and enhance efficiency in collaboration. IndusLink aims to transfer the manual collaboration into digital collaboration.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Faisalabad has emerged as one of the fastest growing city of Pakistan because of its state of the art textile industry. It is the third thickly populated city while it ranked second major in term of revenue generation. Textile is the iconic representation as it alone caters 80% domestic needs of cloths and clothing in addition to contributing around 45% share in total textile export of Pakistan. Its share in GDP of Pakistan is roughly estimated around 22% [\(FCCI, 2024\)](#). As Faisalabad contribute 45% in total textile export revenue of Pakistan, this importance highlights the improvement in textile collaboration is necessary and useful for Pakistan's economy. Still there is no centralized collaborative platform for textile industry. **IndusLink** is centralized B2B collaborative platform specifically designed for textile industry. The primary purpose of this platform is to address challenges and problems that are faced by factory owners and managers in textile industry. Rarely some major factories have all production units in-house, but mostly factories don't have all departments that's why they have to get services from other factories to complete order production. Factories owners and managers visit other factories physically to get required services and for the finalization of agreements. This manual process is time-consuming that leads to delays and miscommunication. Existing B2B platforms are generic and lack with textile sector relevant features. IndusLink will address the challenges by offering the features such as profile creation, create and approve contracts digitally, accessing role-based interface, discussions by integrated real-time chat system, place orders and receive invoices and find specific service provider.

1.2 Aim & Objectives

Aim

Our aim is to build a centralized B2B platform for Faisalabad's textile industry where the manual processes will be shifted into the digitalization. This platform will increase the trust, transparency, efficiency and smooth outsourcing among the textile businesses.

Objectives

- **To Streamline outsourcing & collaboration:** Our platform will reduce the time of outsourcing by at least 30%. This will be achieved by providing the features like AI-based service finder and order management module.
- **To Enable digital contracts & records:** IndusLink will provide digital contracts and will ensure the contracts will be stored digitally and will be looked after approval. This will be achieved by using the Supabase authentication and role-based interface.
- **To Provide service-specific matchmaking:** IndusLink will provide the AI-based matchmaking feature that will at least suggest three best service providers. This will be achieved by implementing rule-based filters.
- **To Improve efficiency & transparency:** Our platform will provide the complete history of orders, each invoice, contracts and chats without data loss this improve the trust and transparency.
- **To reduce the time in manual procedures:** By transforming the traditional manual processes into the digital processes, this will reduce the time and effort of the business owners.

1.3 Problem Statement

1.3.1 Extended Problem

The textile industries in Pakistan lack centralized digital collaborative platform so they face constant challenges. Currently, most factories do not have all in-house units such as knitting, dyeing, cutting, and stitching. Therefore, factories owners visit other factories physically to have collaboration for the finalization of the contracts. The physical visits are time consuming and also leads to miscommunication. There is no a structured system for contract management and orders management so, this creates disputes between the parties, causing delays, losses in finance and inefficiencies in the business. Some existing platforms provides basic product listing, supplier directories and communication features. The existing systems are too generic lack digital contract system, AI-based service matchmaking and secured chat system within one platform. The textile sector of Pakistan export account for over USD 19 billion yearly yet *Digital sustainability in Pakistan's textile sector* (Masooma Zahra, 2025) highlights the lack of digital adoption creates inefficiency.

1.4 Proposed System

Our proposed solution **IndusLink**, provides **centralized web-based B2B** collaboration platform that will replace the inefficient and time-consuming manual procedures of the textile industries by giving the features such as intelligence matchmaking system, Digital contract system, order management, invoice generation, secure chat system, and role-based dashboards.

Firstly, the buyers and the sellers will register themselves and will create their **profiles** in detail such company name, company registration ID, services, and location. The **intelligence matchmaking** will find the reliable partners that will be based on services, production requirements of the buyers, previous contract history and customer reviews.

The core feature of **IndusLink** is **Digital contract system** where both parties can have the contracts digitally. To ensure the transparency the contracts cannot be changed after approval.

The module of **Order management and invoice generation** will allow the users to place order, generate invoices and can have the complete record of all transactions of each contract.

The system will also provide the real time **chat system** where the both parties such as buyer and seller can negotiate, overcome the disputes and can know the progress rate of production procedure after having the contract. Through this system the critical information will be stored within one platform.

In last, the system consist of **role-based Interface** for seller and buyer where both parties can see the related data and tools. For the seller, the data and tools will be the approval of the contracts, client communication, and can get the orders therefore, for the buyer the data and tools will be place an order, review the seller services, seller communication and approval of the contracts.

Our solution will eliminate the physical visits that will reduce the time, effort and cost. It will ensure the trust and transparency by providing structured digital records and will reduce the miscommunication with integrated chat system. Moreover, our system will enhance the competitiveness of Pakistan's textile sector in global by transferring it from manual procedures to digital system.

1.5 Project Scope

Locally Provide Initially our project will be focusing only on Faisalabad's textile industry and will provide the services within Faisalabad's textile industry. Our project will focus on **specific user** such as Manufactures (service provider) and buyers (service seeker). **Web-based platform** Our project will be web-based platform no mobile application.

The **project scope** will include the detailed **profiles** of the users where they can switch their roles as a buyer or a seller, **AI-based service matchmaking, digital contracts, order and invoice management**, and integrated **real time chat**.

Integrated online payment and **National market expansion** will remain **out of scope** in our first phase of the project.

One of the major limitations that can affect our project scope is **Adaption restriction** of the users like transferring their traditional manual processes into the digital process. Other restrictions can be the **internet dependency** where the offline users cannot access the services, and no **integrated online payment**.

1.6 Assumptions & Constraints

Assumptions

- **Digital Literacy:** The users will have the basic digital literacy to use platform.
- **Testing and Feedback:** The initial adopters of our platform will help us in testing to improvement by giving us feedback.

Constraints

- **Restricted geographical Scope:** Initially the focus of our platform is only on the textile industries of Faisalabad.
- **No online payment:** The platform will not have the online payment feature in its first phase.

How these assumptions and constraints are determined:

These assumptions and constraint are determined by focusing on realistic scope of final year project by considering the time and resources and by the analysis of problem statement.

1.7 Social Benefits

- **Economic Growth:** The growth of economy can be beneficial for everyone as more the textile sector will export more it will earn and will grow the economy so, the government by getting the tax can invest more in education, healthcare, and infrastructure.
- **Product Cost decreases:** Smooth collaborations and digital transformation will impact on the production cost and ultimately this will lower the textile products in the local market.
- **Environmental Impact:** Our platform will reduce the physical visits that will lower the carbon emissions from the vehicles. So, this will create pollution free environment for the common person.
- **Women Empowerment:** Our platform will encourage the female's textile business owners to work remotely and will be able to collaborate easily.
- **Community Growth:** Our platform will create the strong local Faisalabad textile community by creating the inter-factory collaboration network.
- **Traditional Shift to Digitalization:** Our platform will shift the old traditional methods in textile business into modern business practices by digitalizing. Also, this increase the digital literacy and the textile businesses will be able to become more competitive globally.

1.8 Business Plan

1.8.1 Business Model Canvas

Table 1.1 Business Model

Key Partners	Key Activities	Value Prop.	Customer Rel.	Customer Seg.
• Factory owners • Industry associations	• Platform development • Customers support	• Collaboration b/w factories • AI-based matchmaking • Secure digital contract system • Ordering system • Real-Time communication	• Feedback and continuous improvement	• Textile manufacturers and buyers
	Key Resources		Channels • Industry trade shows	

	• AI Algorithms • Databases • IndusLink platform	• Role-based interface	• SEO	
Cost Structure • Marketing expense • Deployment & Maintenance expense		Revenue Streams • Freemium Model • Subscription fee		

1.8.2 Problem

In the absence of digital collaborative platform, Pakistan's textile industry is facing challenges for collaboration and suffering from inefficient contract management. Usually, factories do not have complete in-house production units like Knitting, dyeing, printing, etc. Therefore, for completing the production process, business owners and managers visit others factories physically to find the best service provider, for prices negotiation, and contract approval which is firstly time-consuming that leads to delays, miscommunication and secondly there is no standard method to store the business dealings in an efficient way which cause ambiguity and distrust. Current B2B platforms lack textile specific features like smart service matchmaking, digital contracts with mutual approval and role-based interface.

1.8.3 Solution

To address the challenges of textile industry, IndusLink provides a centralized webbased B2B collaborative platform. Business owners or managers can create their profiles as a seller or buyer, by offering smart service matchmaking buyers can easily find the right partners, they can also enable to manage and approve contracts digitally, buyer can also place multiple orders against one approved contract and seller can send order invoices. They can also communicate through an integrated real-time chat system. Single factory can behave as a seller or buyer, by offering role-based interface IndusLink allows to switch roles between seller and buyer to access only relevant features. Above mentioned features will address the textile industry challenges, reduce physical visits, enhance transparency and trust.

1.8.4 Customers

Our primary customers will be the Faisalabad's Textile industry. Factories do not have all production units in-house that's why they look other factories for order completion. They need trustworthy and transparent collaboration, digitally contract management and best way to find the service provider according to their order requirement.

1.8.5 Competitors

- **Alibaba**

A B2B generic platform that connects the sellers and buyers globally. Sellers list their products in bulk and buyers can place large orders. Also, it offers a basic chat system but it lacks textile specific features such as smart service matchmaking, digital contracts approval and role-based interface_([Alibaba, 1999](#)).

- **Fibre2Fashion**

This platform is specifically designed for textile and fashion industry. It offers market insights, product listing and find textile suppliers all over the world but it lacks digital contracts approval, ordering system and integrated chat system ([Fibre2Fashion, 2000](#)).

- **TradeKey Pakistan**

It is a generic B2B platform designed for different industries including textile. It offers product listing, creating profiles, messaging but lacks with digital contracts, smart service matchmaking and ordering system_([Tradekey, 2013](#)).

IndusLink Advantages

IndusLink is industry specific web-based B2B collaborative platform offering AI-based service matchmaking, digital contracts with mutual approval, integrated real-time chat system and most important role-based interface for access control.

1.8.6 Marketing Plan

- **FCCI:** To gain trust and visibility among business owners, collaborate with Faisalabad Chamber of Commerce and Industry.
- **Industry Events and Trade Shows:** Attend exhibition, textile expos and industry events to demonstrate IndusLink value.

- **Search Engine Optimization:** Optimize the platform for keywords like "Industrial digital contract management", "textile collaboration platform".

1.8.7 Revenue

Basic features of this platform are free but to access the premium feature such as AI-based service matchmaking, factories will pay fee monthly. Factories can also pay for advertisement on our platform for the sake of targeting more customers.

1.8.8 SWOT Analysis (Strength Weakness Opportunities Threats)

Strengths

IndusLink is textile industry specific web-based collaborative platform. It offers advanced AI-based service matchmaking to find right partners according to your needs, digital contracts with mutual approval enhance transparency and trust. Also, unique feature that is role-based interface to access relevant features add flexibility for buyers and sellers. Through integrated chat system, they communicate within the platform.

Weakness

AI-based service matchmaking requires high amount of data for accurate results. Initially, building trust and adoption of this platform for new user may be challenging.

Opportunities

In every field of life people want to transfer manual processes into digital processes. That's why business owners need collaboration digitally with other factories to move faster, and compete their competitors. They also want to save their precious time in manual collaboration and put energy in other important concerns of business.

Threats

- **Competition from Existing Platforms:** Some established larger platforms such as Alibaba, Fibre2Fashion, TradeKey Pakistan may introduce the textilespecific services in the future.
- **Economic Uncertainty:** The up and down in the economy condition of Pakistan may reduce the investment in the digital adaption.
- **Infrastructure Condition:** Bad internet services and electricity instability in some areas may affect the use of our platforms.

1.8.9 FAB Analysis (Features, Attributes, Benefits)

Table 1.2 FAB Analysis

Feature	Attribute	Benefit
Sellers & Buyer Profiles	Services and infrastructure detail	Engage more customers by showing their abilities credibility's.
AI-based Matchmaking	Smart suggestion feature	Saves time by suggesting the best fit service provider instead of typical searching.
Digital Contract System	Online create, approve & protect contracts securely	Enhance transparency by eliminating physical visits & paper-based agreements.
Ordering System	Place orders & generate invoices	Complete history of all orders against single contract enhance visibility.
Integrated Chat System	Real-time communication	Discussions about prices, contracts, etc stored within the platform that minimize verbal communication, miscommunication.
Role-based Interface	Switch roles between buyer & seller	Access only relevant features and data which increase flexibility.

1.9 Report Layout

This chapter covers introduction of our IndusLink, highlighting the problem statement, proposed solution, project scope, social benefits and business plan. Further the chapters will cover the Literature review, Requirement Analysis.

CHAPTER 2: BACKGROUND AND EXISTING WORK

2.1 Background

Most factories do not have the all-in-house units such as knitting, dyeing, cutting, stitching and packaging. Therefore, factories owners and managers visit other factories physically to have collaboration for the finalization of the contracts. The physical visits are time consuming and also leads to miscommunication. There is no a structured system for storing contract and other business data so, this creates disputes between the parties, causing delays, losses in finance and inefficiencies in the business. As Faisalabad is the hub of textile sector and improvement in this sector by digitally can increase the efficiency in the process that can improve the overall economy of the country.

In 1999 *Alibaba* (Alibaba, 1999) that made in China enabled online supplier-buyer connections, but the main focus was on product listings. In 2000 *Fibre2Fashion* (Fibre2Fashion, 2000) was launched for textile sector and their focus was on market insights and product directories. In 2010 globally the textile industries grew rapidly and began to adopt ERP systems, and automation but in Pakistan still the process was manual using traditional methods that highlights the importance of digital transformation.

2.2 Literature Review

The literature review's purpose is to justify IndusLink, B2B collaborative platform for textile sector by combining the academic and industrial research. The review focuses on the existing research and system that are based on B2B marketplaces, digital transformation in textile sector, that are relevant to the scope of IndusLink.

The literature review is organized into the following subheadings: **(1)** Digital Transformation in the textile industry **(2)** B2B platforms & their limitations for Textile industry **(3)** Technological Efficiency can improve the production process.

2.2.1 Digital and Sustainable Transition in Textile Industry

Several articles highlighted the importance of need for digital transformation in the Textile sector of Pakistan. One of them is *Digital sustainability in Pakistan's textile sector (2025)* (Masooma Zahra, 2025) their focus is on the low adoption of digital

platforms, the study shows the technological adoption can improve the efficiency, the key predictors are performance, innovation cost, social influence, lean thinking.

- **Analysis:** The studies shows that digital transformation can improve operational outcomes. So IndusLink will fulfil these requirements.
- **Strength:** This study highlights a current and critical problem, the digital transformation in textile industry of Pakistan and this sector is a core economic sector.
- **Weaknesses:** This study have too many statistical terms without any simplification or summarization of practical meaning.

2.2.2 B2B Platforms and their Limitation in the Textile Industry

Some digital platforms such as Alibaba, Fibre2Fashion, TradeKey provide the services for the industry such as product listing, supplier discovery, and real time chat but they are generic and they do not contain all features for the textile industry that are essential for the outsourcing.

- **Analysis:** Existing platforms provides the features like product listing, supplier discovery but do not have the specific features for textile industry as AI-based matchmaking and contracts for inter-factory collaboration. So IndusLink will fulfil these gaps by providing the missing features.
- **Strength:** These platforms provide the international buyer and seller connections, market insights and supplier directories.
- **Weaknesses:** These platforms lack AI-based service matchmaking and digital contracts that are essential for outsourcing.

Reasons of Project (In the light of literature)

- The literature's studies show there is low adaption of technology and there must be a digital system for textile industry.
- The generic existing platforms lack the textile specific features.
- The adaption of digital system in textile industry can enhance the efficiency, transparency, and trust that will able the textile sector of Faisalabad to stand in the global market more effectively.

- The digital system for textile industry will replace the manual processes that will improve the delays and disputes.

Selection Criteria of the Project

- **Directly Related to Industry Needs:** As we have identified the industry related problem, so system must be designed to fulfil the needs of industry.
- **Simplicity and Usability:** The interface of IndusLink must be simple, flexible and user friendly, so the non-technical persons will able to use easily.
- **Security and transparency:** The order, invoices, contract, and chats should have to be authenticated and the data must be stored within the platform.
- **Scalability and feasibility:** The project will be supported for the future integration such as online payment within the system and also the project must be feasible within the deadline.
- **Low-Cost:** The platform is the student project so we will use the open-source technologies to maintain the cost while we will ensure the performance of the system.

Key features of IndusLink

- **Detailed Profiles:** The detailed profiles of the buyer and seller will be constructed.
- **Intelligence Matchmaking:** AI will find the best service provider for the buyers according to their requirements.
- **Digital contracts:** Digitals contracts will be provided within the platform to enhance the transparency.
- **Order Management:** The platform will provide the order management system where the user will be able to place orders and generate invoices.
- **Chat System:** Secure chat system will provide to the buyer and seller for discussions.
- **Role-Switching Interface:** The user will be able to switch his role as a buyer or a seller with the same account.

Functional Requirements:

- User Creation/Authentication

- Profile Management
- Role-Switching Interface
- AI-based Matchmaking
- Contract Management
- Order Management
- Real-Time Chat System

Non-Functional Requirements:

- Security
- Performance
- Usability
- Data Integrity

2.3 Literature Summary

The low adoption of digital transformation, and its importance in the industry to improve the efficiency of the production process. Some B2B platforms do not have the textile industry related specific feature for the collaboration. Technological adoption can improve the delays, disputes and transparency by providing the digital contracts, order history and real time communication. IndusLink will fulfil the gaps by provide the AI-based matchmaking service-provider, digital contracts, order management, invoice generation and real-time chat system for inter-factory collaboration.

CHAPTER 3: REQUIREMENTS ANALYSIS

3.1 Stakeholders List (Actors)

Table 3.1 Stakeholders List

Students Development Team	Responsible for the development of Final Year Project IndusLink .
Supervisor	Guide us throughout the project, ensure project align with academic and industry requirement.
Evaluators	Evaluate our project and provide feedback recommendations and give marks at the end.
End users	After the launching of IndusLink in textile industry, they will use it.

3.2 Requirements Elicitation

It is process of identifying and understanding the needs of stakeholders.

Techniques:

Techniques to elicit the requirements are as follow:

- **Discussions with Factory owners:** Discussed with business owners and managers to understand the inefficiencies and pain points in manual processes and collaboration.
- **Existing System's Analysis:** Analysed the existing systems like Fibre2Fashion, TradeKey Pakistan to understand the gaps and absent features.

3.2.1 Functional Requirements

- **User Creation/Authentication**
Users can register as a seller or buyer and access only relevant interface. If seller create his account so he will be able to access the buyer side without creating the buyer account and if buyer create his account and want to access the seller side so he has to create the seller account as well. To access the platform functionalities, user have to log in.

- **Profile Management**
Sellers can create profile with services, company name, location, etc and also update their business profile whenever they want to change it.
- **Role-switching Interface**
Single Factory can behave as a seller or buyer so that they can switch their role between seller or buyer to access only relevant interface with relevant features.
- **AI-based Service Matchmaking**
System will recommend the most suitable factory based on location, prices, services and according to your past preferred factories.
- **Contract Management**
Users can create new contract and approve it securely and system will maintain all the history of contracts. Once the contract is approved it cannot be altered.
- **Order Management**
Buyers can be to place many orders against single contract and seller can generate invoices, then send it to buyer and both can view history.
- **Real-time Chat System**
Both buyers & sellers can communicate through integrated chat system in real time.
- **Notifications**
System can send alerts on contracts approval, messages.

3.2.2 Non-Functional Requirements

- **Security**
Ensure proper authentication and secure storage for contract and business data.
- **Performance**
System should handle more request at a time and give response in minimal time.
- **Usability**
User-friendly interface for both technical & non-technical users for IndusLink.
- **Data integrity**
Ensure that data entered by user must be accurate.
- **Scalability**
System must support future expansion.
- **Maintainability**
Application modules should be documented and modularized for easy updates.

3.2.3 Requirements Traceability Matric

Table 3.2 Requirement Traceability Matric

Requirement Description	Source	Goal
User Authorization & Authentication	Students, Supervisor	Secure access and rolebased control
Profile Management (Factories & Buyer)	Textile Business Owners	Service visibility
Profile Switching (Buyer or Seller)	App Users	Flexible for user to switch the role as a buyer or seller
AI-based Service Matchmaking	Buyers	Find the most suitable partners
Digital Contract Management	Business Owners	Reduce physical visits and delays
Order & Invoice Management	Buyers/Sellers	Manage orders and ensure financial transparency
Real-time Chat System	Buyers/Sellers	Improve communication and reduce disputes
Notifications	Users	Keep users updated about activities
Performance	Supervisor	Ensure smooth system with good response time

Usability	Non-technical users	Easy adoption for non-technical users
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3.3 Use Case Design/Description

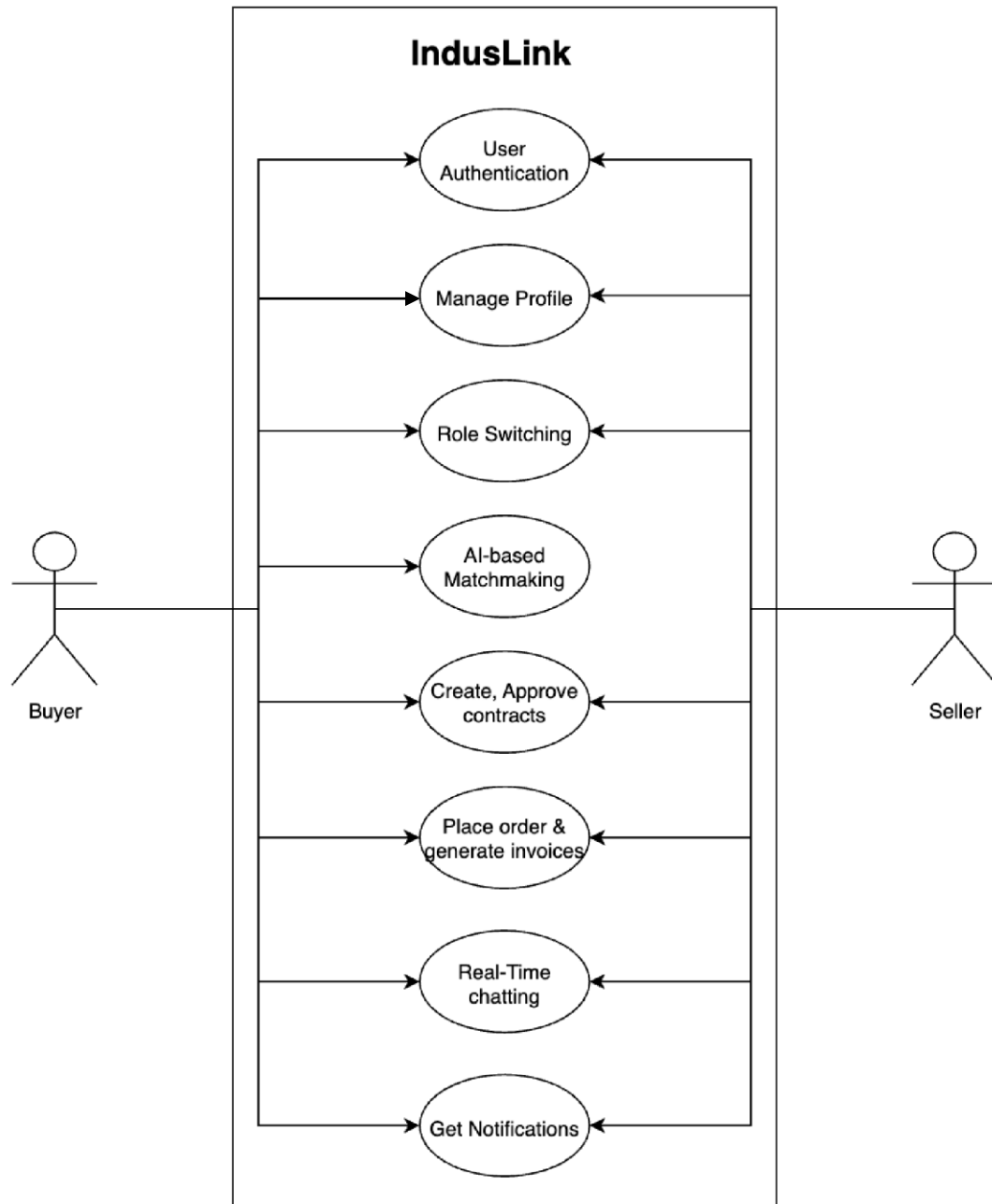


Figure 3.1 Use Case Diagram

This Figure 3.1 show the use case diagram of IndusLink system. This diagram represents the main actors and interaction with the system.

3.4 Software Development Life Cycle Model

For development of IndusLink we adopt **iterative model**.

Justification: By using this approach, system is developed in manageable iterations and also iterative model allow continuous feedback from stakeholders, help to refine the system modules time to time and flexible for adding further features.

3.5 Specific Requirements (Hardware and Software Requirements)

3.5.1 Hardware Requirements

Table 3.3 Hardware Requirements

Requirements	Specification
Processor	2.6 GHz Quad-Core Intel i5 / i7
RAM	Minimum 8 GB but recommended 16 GB
Storage	20 GB free space
GPU	Sufficient integrated GPU

3.5.2 Software Requirements

Table 3.4 Software Requirements

Requirement	Detail
Operating System	Windows 11, macOS 10.15 +
Next JS	Frontend Framework 15.4.6
JavaScript	Programming Language ES6
Tailwind CSS	4.1.12
Python	Programming Language 3.12.5

FastAPI	Backend Framework 0.115.0
Supabase	Database 2.55.0
IDE	Visual Studio latest version
Browser	Chrome, Safari

CHAPTER 4: SOFTWARE DESIGN SPECIFICATION

4.1 Design Models

The approach chosen to design IndusLink is Modular Design in which the system is divided into different independent and grouped modules. Each module is built for specific functionality such as profile management, contract management, orders management, communication and etc. By using this approach, it is easy for development team to add more features in future, test each module efficiently and maintain whole system over time with less efforts.

The chosen diagrams include Block Diagram, Software Architecture Diagram, Entityrelationship Diagram, Data Flow Diagram, Hierarchical Diagram, Flowchart, Sequence Diagram and Activity Diagram. Block and Software Architecture diagrams defines the system layers and components. ERD and DFD show the data and process flow between different modules. Sequence, Activity and Flowchart Diagrams highlight how different modules interact during system operations.

4.2 Work Breakdown Structure

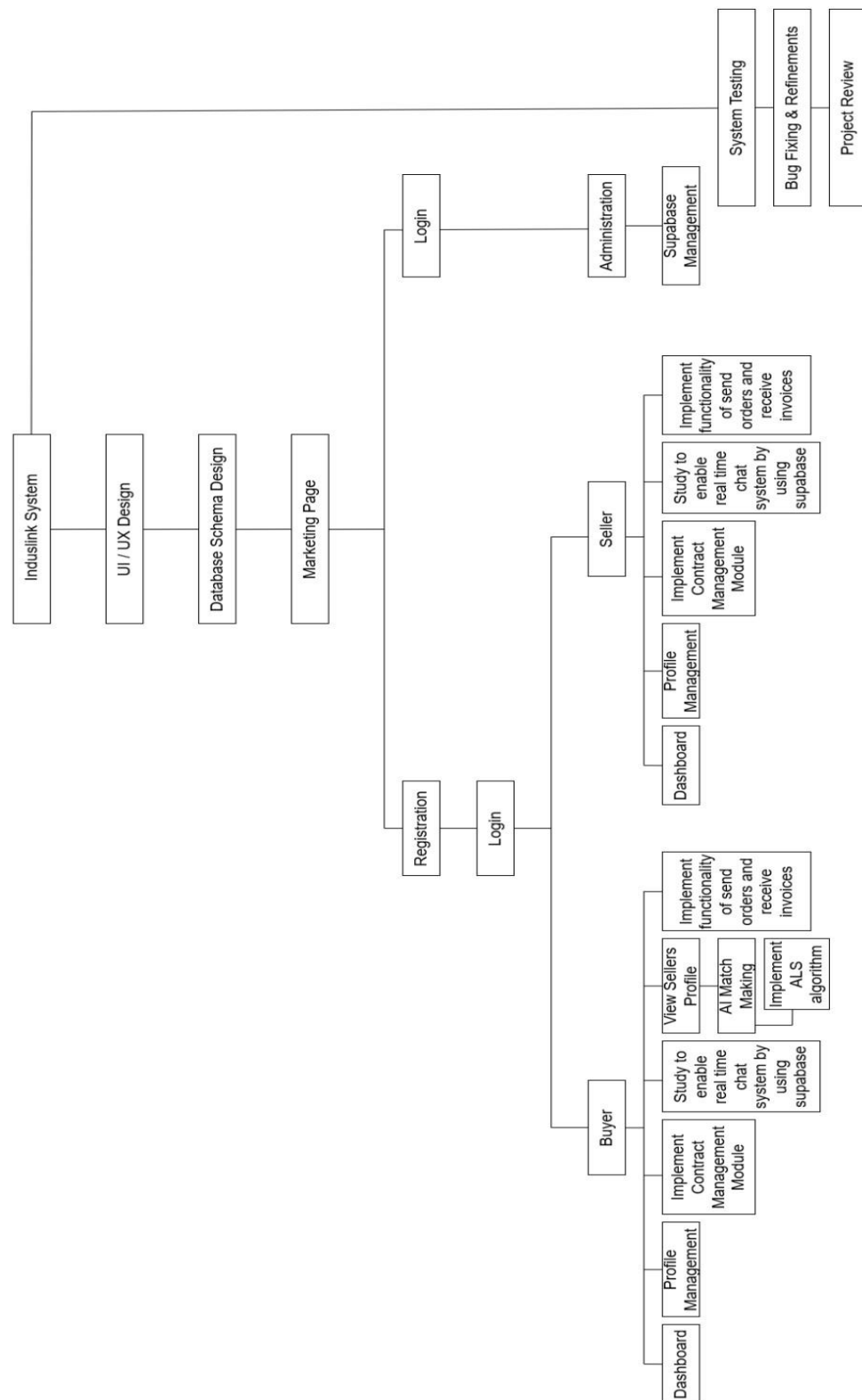


Figure 4.1 Work Breakdown Structure

Figure 4.1 illustrate the work breakdown structure of IndusLink. It show three main divisions one is seller division, second is buyer and third is Admin.

4.3 System Architecture

- **Block Diagram**

This Diagram represents higher-level main workflow of overall system. In IndusLink, it is very important to show the overall main components of system and also to show how different roles like buyers and seller interact with the system. The block diagram mainly the high-level structure which provide help us to understand the complete structure and workflow of system .

Component Diagram

Component diagram shows how different modules interact with system and what is the relation of one module to other and how they connect with each other. Mainly in IndusLink, there is two main sections the buyer and seller and their further components and their interaction is defined in each section.

- **Software Architecture Diagram**

Software Architecture Diagram shows main structure of system in form of four different layers. It shows the interaction and flows of the components and help us to understand the design of the system in better way.

4.3.1 Block Diagram

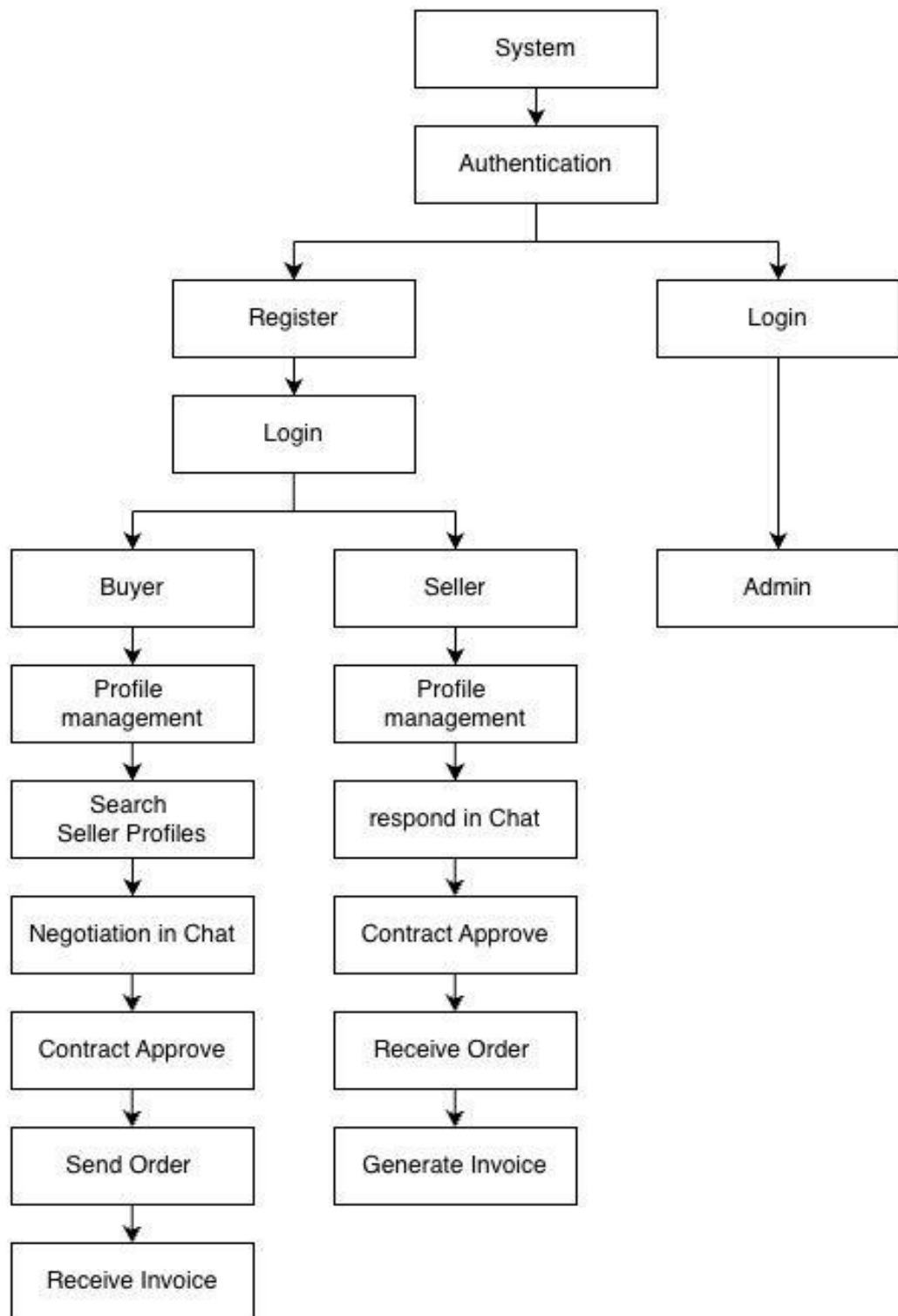


Figure 4.2 Block Diagram

Figure 4.2 show the IndusLink system in block diagram. This diagram show the high level workflow of IndusLink.

4.3.2 Component Diagram

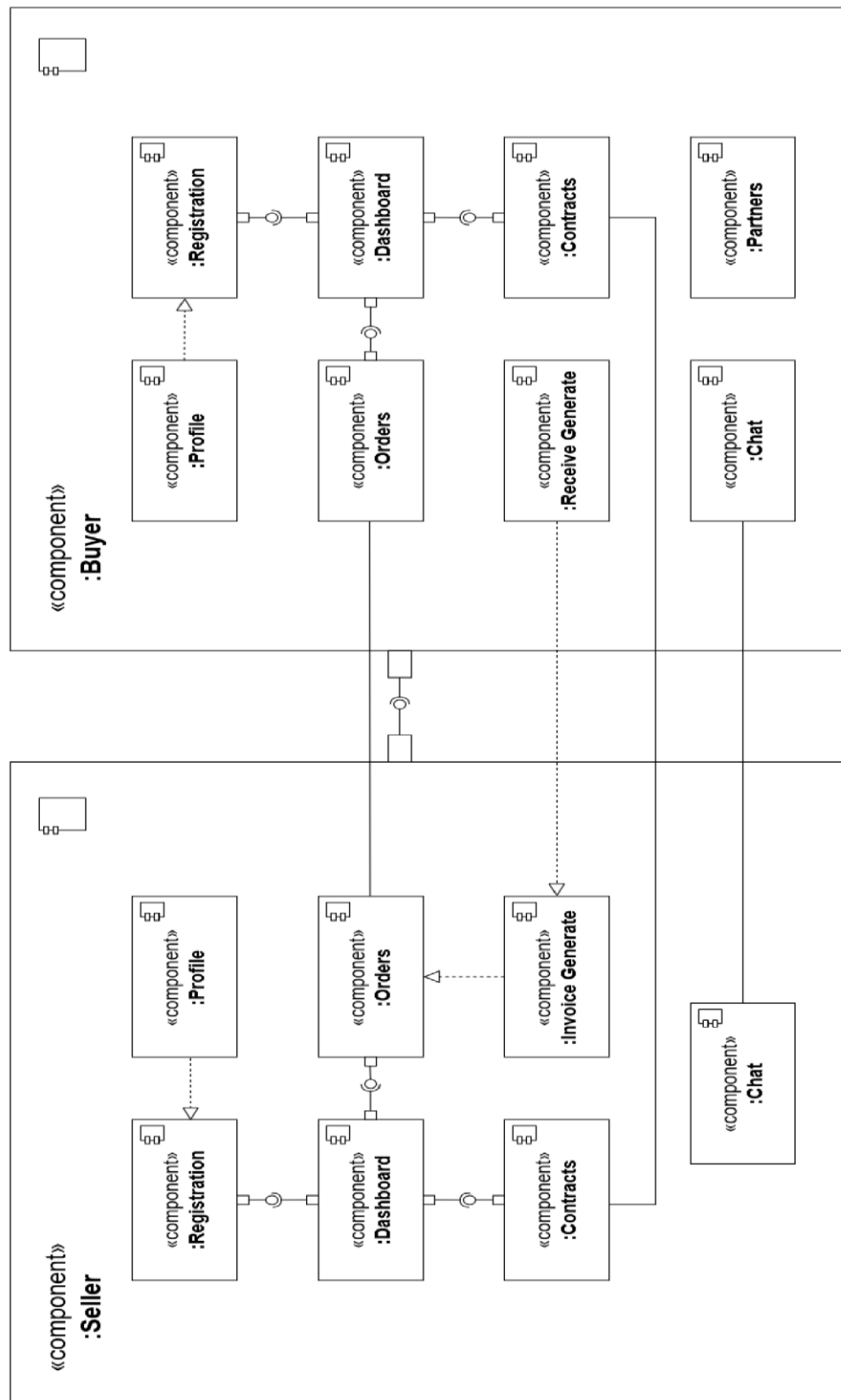


Figure 4.3 Component Diagram

Figure 4.3 shows how different modules will interact with each other.

4.3.3 Software Architecture Diagram

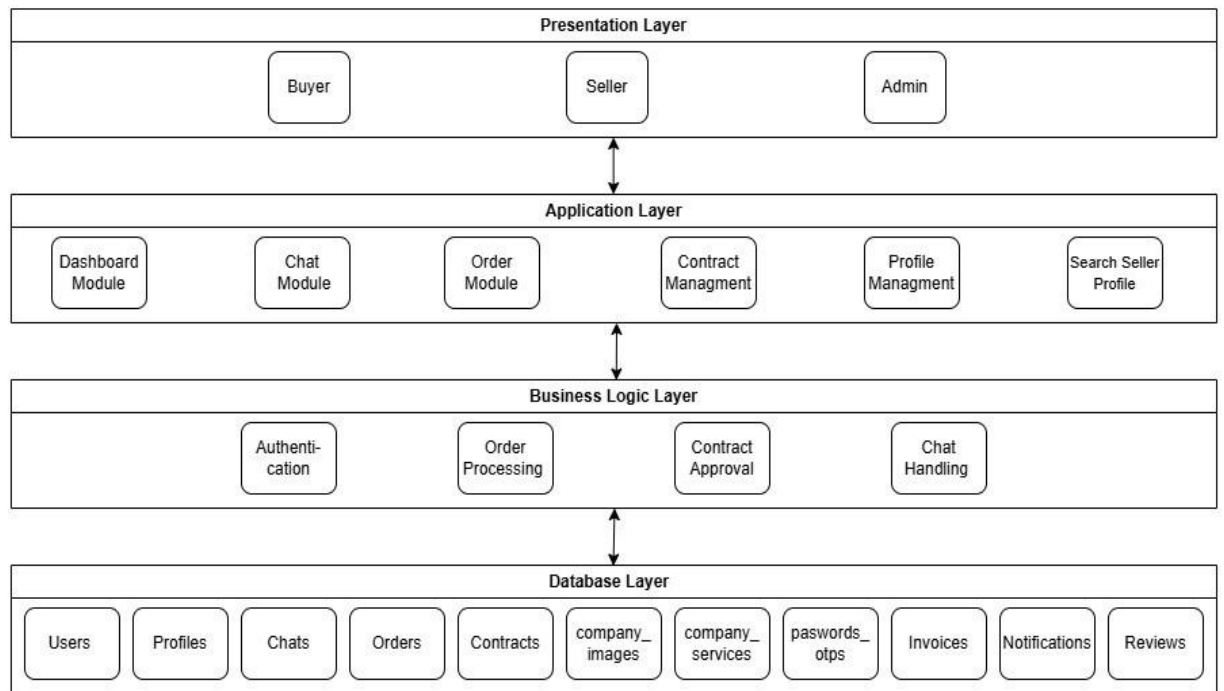


Figure 4.4 Software Architecture Diagram

Figure 4.4 shows how different layers of the system will interact to each other for the data correct flow.

4.4 Data Representation [Diagram + Description]

- **ERD**

Erd helps us to understand the clear structure of system. Erd shows relationship between entities & help developers in planning before implementation. It shows the attributes of each entity and reduce the redundancy in system components.

- **DFD**

DFD is use to show internal flow of system and how one module of system interact and communicate with another module. In IndusLink, this helps us to understand the overall working of system and shows the flow of user roles, major processes, data storage in the system.

- **Hierarchal diagram**

Hierarchical Diagram is used to show tree like structure of system. In IndusLink, It helps us to visualize the main components and their top-down view of the system. This shows the parent child relations and high lights the role of the users.

4.4.1 Entity-Relationship Diagram (ERD)

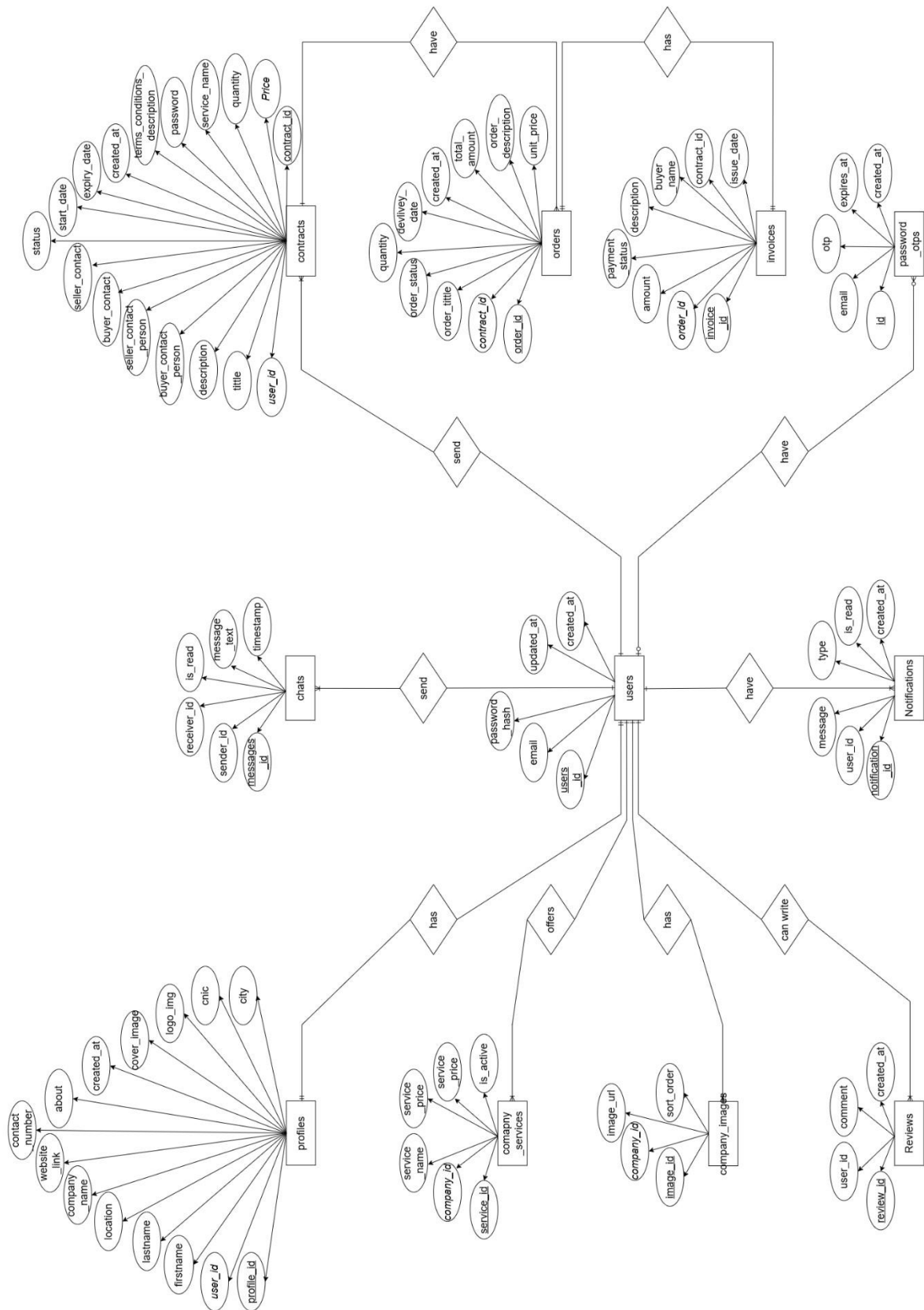


Figure 4.5 ERD Diagram

Figure 4.5 show the entities and their attributes of IndusLink.

4.4.2 Data Flow Diagram (DFD)

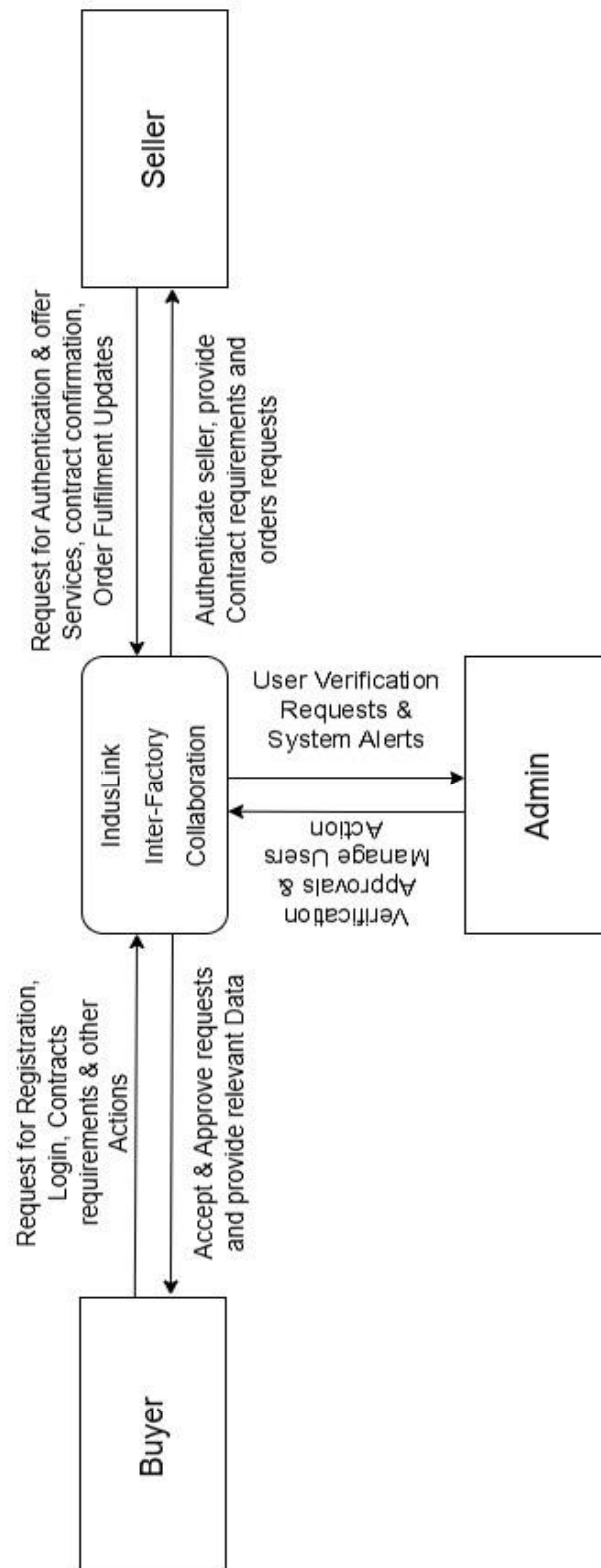


Figure 4.6 Data Flow Diagram Level 0

Figure 4.6 explains the basic level system architecture just to get the main idea.

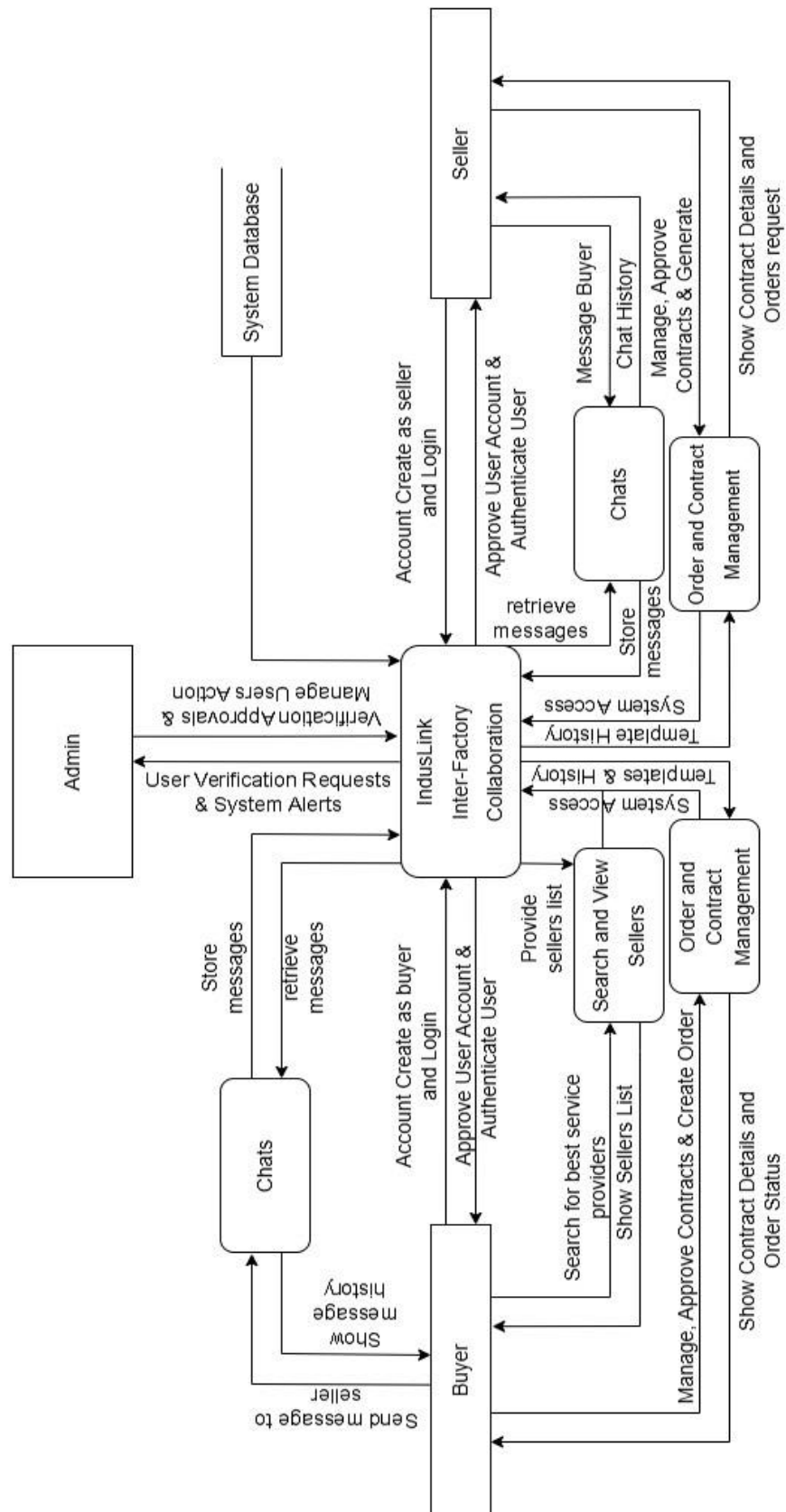


Figure 4.7 Data Flow Diagram Level 1

In Figure 4.7, highlight the major processes and interaction of system.

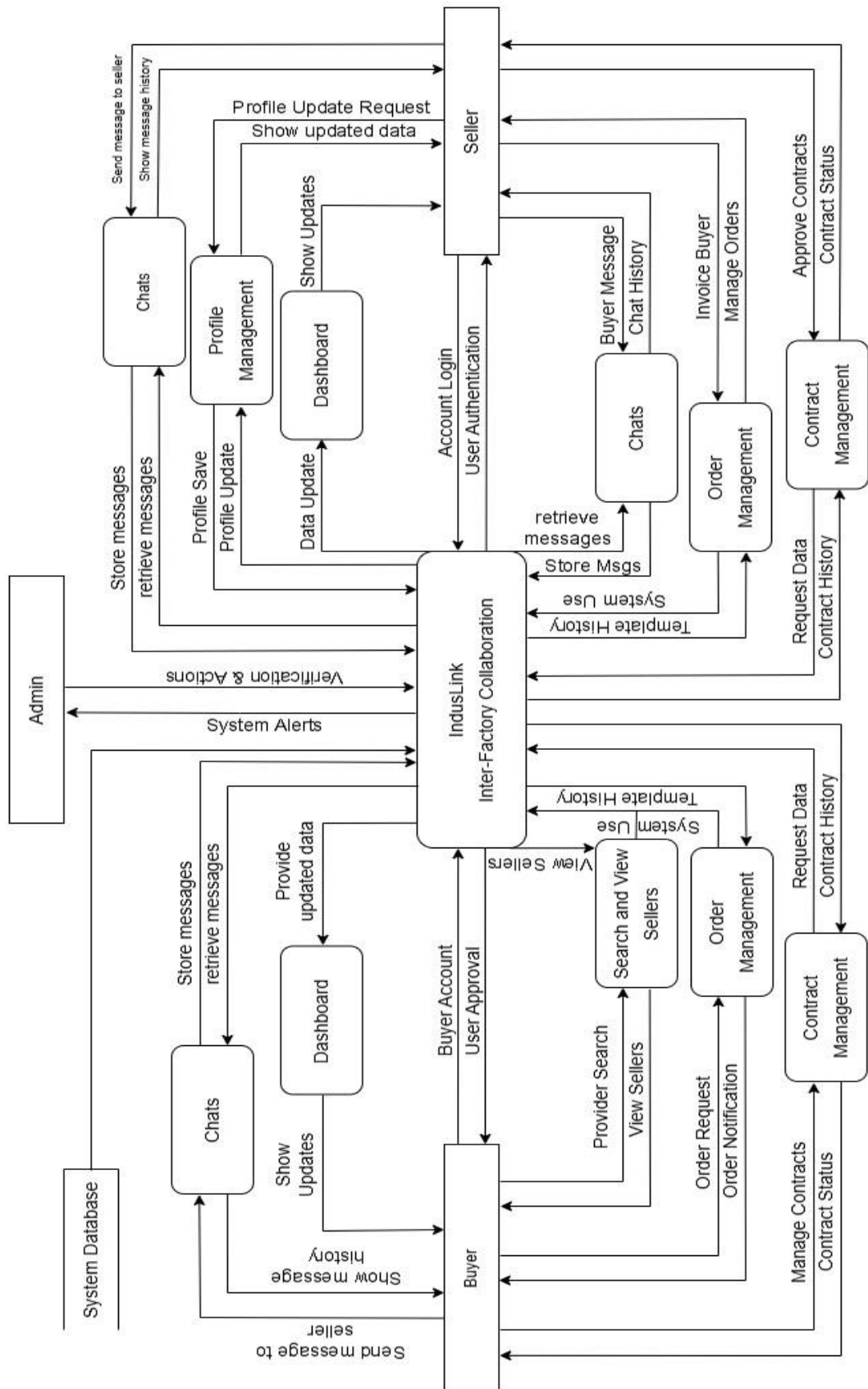


Figure 4.8 Data Flow Diagram Level 2

Figure 4.8 represents the all functionalities and their interaction with depth.

4.4.3 Hierarchical Diagram

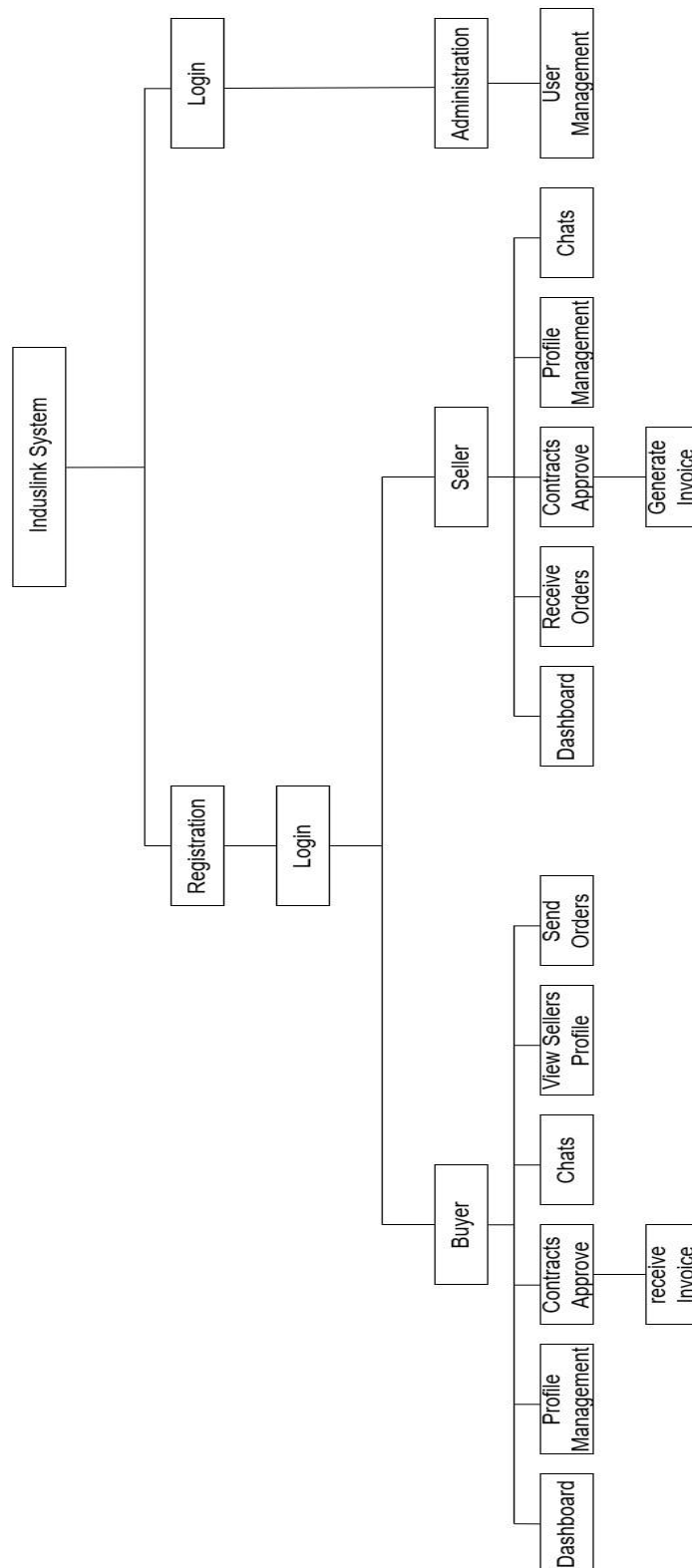


Figure 4.9 Hierarchical Diagram

Figure 4.9 show the main components and their interaction for simple understanding.

4.5 Process Flow/Representation

- **Flowchart Diagram**

It is the process diagram that shows the easy to understand decision-making step-by-step process in IndusLink. The main symbols of flowchart are rectangles, diamonds and arrows.

The main significances of flowchart that influence IndusLink are optimization of processes, better user experience design and identification of risk.

- **Sequence Diagram**

It is UML interaction diagram that illustrates objects interaction in IndusLink with each and other in a sequence. It shows step by step interactions procedure between all the components. The main components of this diagram are Lifelines, Activation bars, Message, Return arrow and Loops.

The significances of this diagram are blueprint for the technical implementation, guide for the optimization and planning for integration.

- **Activity Diagram**

This diagram is UML behavioural diagram that shows workflow of B2B collaboration platform, IndusLink. It shows the interaction of actors with system and uses swim lanes to separate the activities of buyers and sellers. The main components of Activity diagram are Decision, Activities, Start/End node and flow control.

The main significance of activity diagram that influence on IndusLink are, it clarifies the workflow, help to identify system requirements and it supports the design of system.

4.5.1 Flowchart

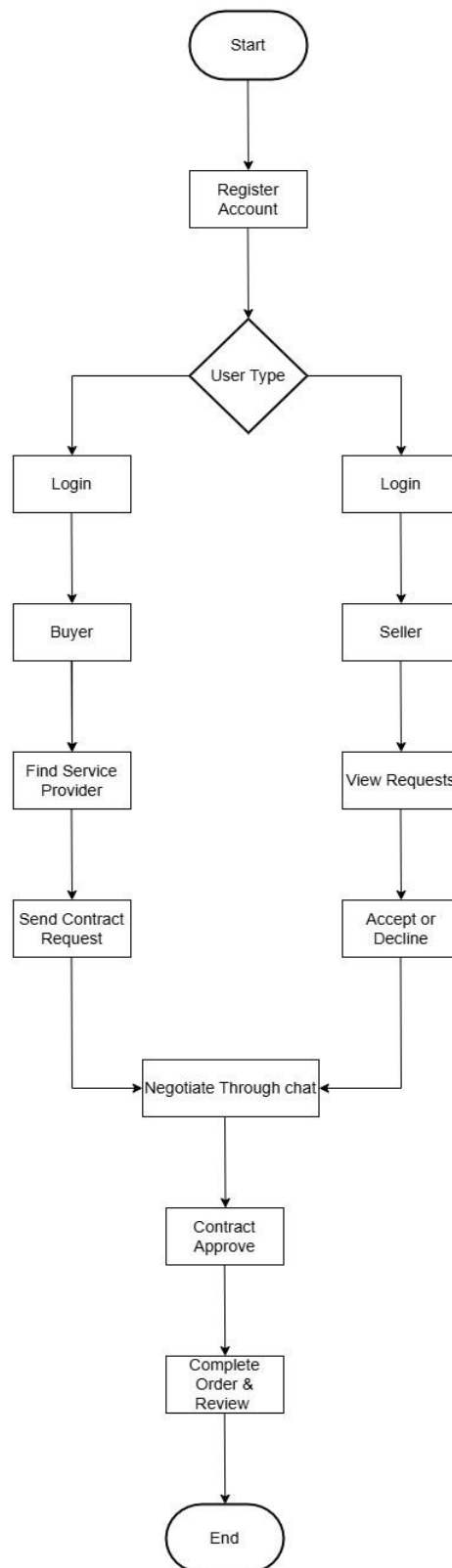


Figure 4.10 Flowchart

Figure 4.10 highlight the step by step process of system for better understanding.

4.5.2 Sequence Diagram

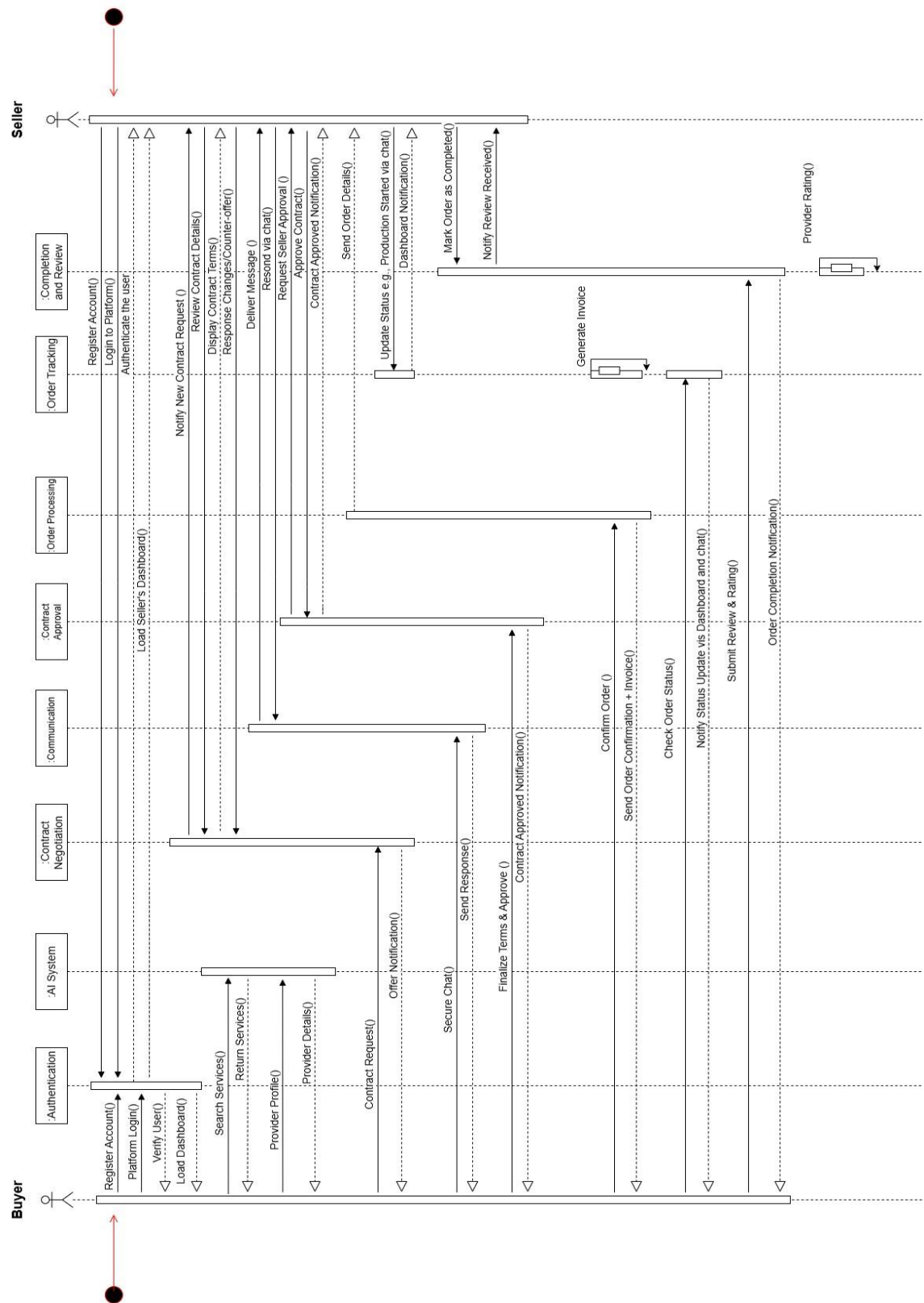


Figure 4.11 Sequence Diagram

Figure 4.11 show the interaction, their timelines and loops visually.

4.5.3 Activity Diagram

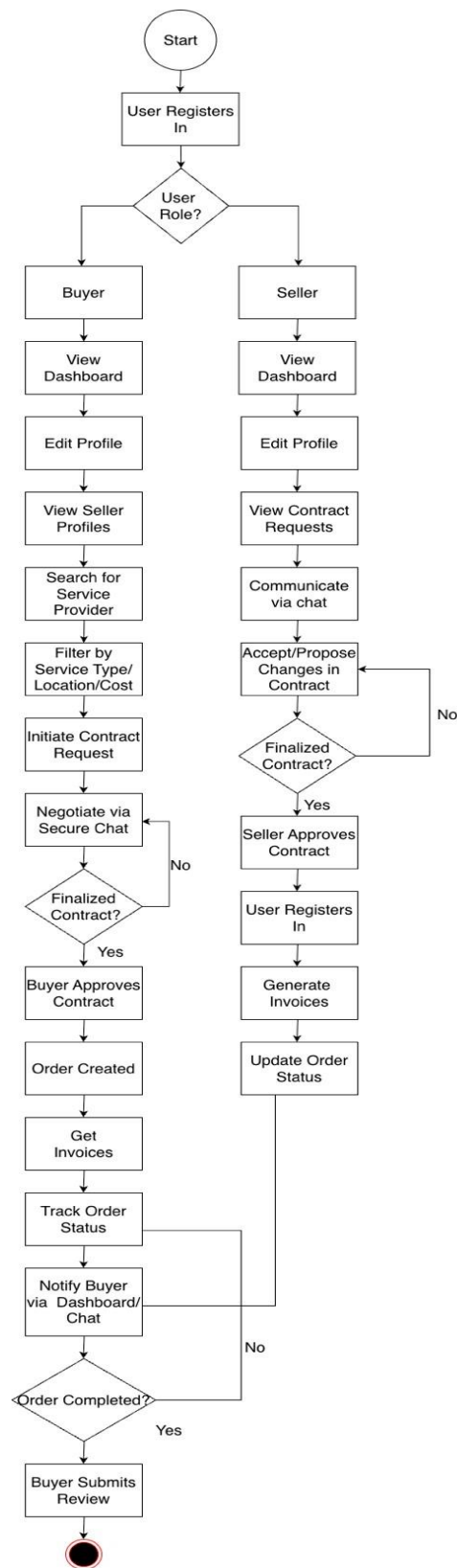


Figure 4.12 Activity Diagram

Figure 4.12 highlights the major workflow of IndusLink.

CHAPTER 5: IMPLEMENTATION

5.1 Algorithm

1. User Registration Algorithm

This module gets the required data from user like his personal details, company information, etc and validate the information. After validation system stores the data in database. This allows the businessmen to create their profiles on IndusLink to use features of this platform.

Algorithm Steps

1. Users click on **Register Now** button.
2. System redirect user towards registration page.
3. Users enter all required data and info.
4. System validates the details.
5. If user exists in database, system show the error message.
6. If user doesn't exist in database, system create a new user successfully.
7. Redirect to login page.

Pseudo Code

Click on **Register Now** button.

```
INPUT firstName, secondName, emailAdress, phoneNumber,  
CNIC allValid = isFirstNameValid && isLastNameValid &&  
isEmailValid && isPhoneValid && isCNICValid
```

```
IF allValid THEN
```

```
    INPUT companyName, location, country, city,  
websiteLink, logoImg, mainImg allValid = islogo &&  
isMainImage && isCompanyValid
```

```
    && isLocationValid && isWebsiteValid;
```

```
    IF allValid THEN
```

```
        IF toggleBtn == true
```

```
            INPUT serviceName, servicePrice, serviceLogo
```

```
            User = seller
```

```

ELSEIF toggleBtn == false

    User = buyer

    ELSE

        DISPLAY "Fill all fields data"

        INPUT aboutCompany      isValid=
isCompanyAbout

        IF isValid THEN

            INPUT password,confirmPassword

            isValid=isPassword, isconfirmPassword

            IF isValid THEN

                INPUT companyImages,
companyDescription

            ELSE

                DISPLAY "Fill all fields data"

        ELSE

            DISPLAY "Fill all fields data"

    ELSE

        DISPLAY "Fill all fields data"

Redirect to Login Page

```

2. User Login Algorithm

Login module is responsible for authentication. It takes the user email and password then validate it with system's stored records. If record is correct, user is authenticated otherwise system show an error message. Main purpose is to give access only to verified and authorized users.

Algorithm Steps

1. Users enter his/her email.
2. Then enter his/her password.
3. Now user click on login button.

4. System validates user email and password with stored email and password.
5. If it is wrong, system show an error message.
6. If it is correct, user successfully logged in to system.
7. If he/she forget his password, he clicks on forgot password link.
8. Then enter his email.
9. Again system validates the email.
10. If it is matched, user gets otp on his email and enter it.
11. Now system validates the otp.
12. If it is correct, allow the user to create new password.
13. If otp not correct, system show error message to enter correct otp.

Pseudo Code

```
INPUT Email, Password

IF AuthenticateUser THEN

    IF hasServicesData THEN

        Redirect to sellerside

    ELSE

        Redirect to BuyerSide

ELSE

    DISPLAY "User Not Found. Login Failed"

IF user Click on Forgot Password

    INPUT email

    sendOTP =OTP send to email

    INPUT OTP

    If OTP == sendOTP

        User can reset password.

    ELSE

        DISPLAY "Failed to reset Password"
```

3. Profile Management Algorithm

This allows the authorized users to view and modify their profiles on IndusLink. Before saving, editing details, system ask for confirmation then update in database. This helps the users to keep their profile data accurate and up-to-date.

Algorithm Steps

1. After login as seller redirect to seller's dashboard, same for buyer.
2. When user click on profile button from side panel he navigates to Profile Management Page.
3. System retrieves all user profile data and show it.
4. User can edit company location, website link, cover image, services, about and gallery images, system ask for confirmation then update it.
5. Same for buyer with his data.

Pseudo Code

```
DISPLAY profile data
```

```
IF USER == BUYER
```

```
    DISPLAY coverImg, companyLogo, companyName,  
    companyAdress, companyWebsite, aboutUs, gallery
```

```
    IF user CLICK on coverImg, companyAdress,  
    companyWebsite, aboutUs, gallery THEN
```

```
        Can EDIT fields
```

```
IF USER == SELLER
```

```
    DISPLAY coverImg, companyLogo, companyName,  
    companyAdress, companyWebsite, aboutUs,  
    gallery, services
```

```
    IF user CLICK on coverImg, companyAdress,  
    companyWebsite, aboutUs, gallery, services THEN
```

```
        Can EDIT fields
```

4. Real-Time Chat System Algorithm

Chat system facilitates the buyers and sellers to communicate immediately. System allows the users to communicate in real time. System validates, message is not empty then stored in database with time stamp. If recipient is offline, system show messages when recipient is online.

Algorithm Steps

1. User clicks on chats button from panel.
2. Chat interface is opened for user.
3. User can start conversation to buyer as seller or to seller as buyer.
4. User types a message in input box.
5. Then click on send btn.
6. System stored messages in database with timestamp.
7. Then send message to user in real-time.
8. User can read messages if he logs in.

Pseudo code

Click on chat button

IF newUser THEN

 IF newUser == Buyer THEN

 DISPLAY seller profile to initiate chat

 ELSE

 DISPLAY buyer profile to initiate chat

ELSE

 DISPLAY all previous chat with their companies

Click on company to send message

INPUT message

Then click on send btn to send message IF

validInput

 Send notification to the receiver

5. Contract Management Algorithm

Contract management is responsible to create and manage contracts between buyers and sellers. It ensures the agreement is properly created and stored within the system.

Algorithm Steps

1. Buyer see sellers profiles and click on **start contract** button.
2. After clicking on start contract button it navigates the buyer to contract interface.
3. That seller is added in contract interface.
4. If buyer clicks on that seller company, buyer moves to next page.
5. Buyer clicks on new contract button to initiate the contract and fill out required details and click on done button.
6. Now contract is shown to that seller with filled data.
7. Now seller add his details and done from his side.
8. System updates the contract to buyer.
9. If buyer wants to make any changes click on edit button and save changes.
10. Now buyer click on approve button.
11. Buyer enter his password.
12. System validate the password with stored password.
13. If it is correct, system send otp to buyer.
14. Buyer enters otp then system verifies the otp.
15. If it is correct contract will approved from buyer side.
16. Now seller can approve contract with the same process after buyer approval.
17. If both seller and buyer has any query, can use contract integrated chat system.

Pseudo code

Buyer create new Contract

Input phoneNumber, contactPerson, serviceName,
measuringUnit, paymentTerms Click done to send
contract to seller

Seller receives Contract

Input phoneNumber, contactPerson, price,
terms_and_policies

```

Click done to update contract for Buyer

IF changes required THEN

    CLICK edit button

Else

    CLICK approve button

Seller CLICK approve button after approving from buyer.

INPUT password IF validPassword

sendOTP=Send OTP to the user email

    INPUT OTP

    IF OTP== sendOTP

        DISPLAY "Contract Approved"

ELSE

    DISPLAY "Error Message"

For Contract Negotiation use Contract chat system

```

6. Order Management Algorithm

Contract management is responsible for to create orders and invoices against specific contracts while maintaining the records of previous orders.

Algorithm Steps

1. This module show only the companies with which buyer has at least one contract.
2. Buyer click on a company.
3. Now Buyer first select the contract, then he will be able to create an order.
4. After the order creation, it will showed on seller's page.
5. If seller accept that order, then the order's status updated and invoice will generate.
6. Invoice will be show to buyer.
7. Then order status updated whenever seller update it like production, dispatch etc.
8. After dispatch buyer will click on mark as delivered button for confirmation.
9. Now order is completed on both sides
10. No-one can delete or edit it, just able to view.

Pseudo code

```

IF USER == Buyer

```

```

    DISPLAY contracted factories

    IF user CLICK on any factory
        DISPLAY all the orders Data

        IF user CLICK on new Order
            Select services, quantity

            Click Create Order and send to seller

        IF user CLICK on view invoice
            Can view the invoice send by the seller

IF USER == Seller

    DISPLAY all the factories who sent orders

    IF user CLICK on any factory

        DISPLAY all the orders

        IF user receive order
            Click view button to view the order

            User click on accept button, generate
invoice and send to buyer

            User can update the status of order

```

6. AI-based Service Match-Making

This will show factories according to user's requirements instead of user inspect each profile.

Algorithm Steps

1. User click on AI search button.
2. Give a prompt to AI according to his requirements.
3. AI will process the prompt, then show filtered factories to user while providing the reason.

Pseudo code

```

IF USER CLICKS on AI search button

INPUT prompt

```

AI process the INPUT

Display Filtered Factories

5.2 External APIs

- **Supabase API**

Supabase is an open-source **backend-as-a-Service** platform so, developers doesn't need to handle backend code from scratch. It uses PostgreSQL. To handle backend services such as database operations, authentication and realtime communication, project integrated with Supabase API. System can store and manage user data securely by using Supabase API.

- **Purpose**

To create tables, we just run SQL query directly in Supabase SQL editor. Database CRUD operations can be easily performed so that data storage and retrieval can be done efficiently. The other main purpose is real-time communication instead of using web sockets, real-time communication can be done by enabling Supabase real-time feature.

- **Data-Retrieval & Communication with API**

In Supabase the build-in methods of JavaScript SDK like `.from()`, `.select()`, `.insert()`, `.update()`, and `.eq()` is used and they directly work on the PostgreSQL tables like users, profiles, orders, contracts, and chats etc. Data retrieve through Supabase API where application fetch information from database and receive response in JSON format and display on the frontend interface.

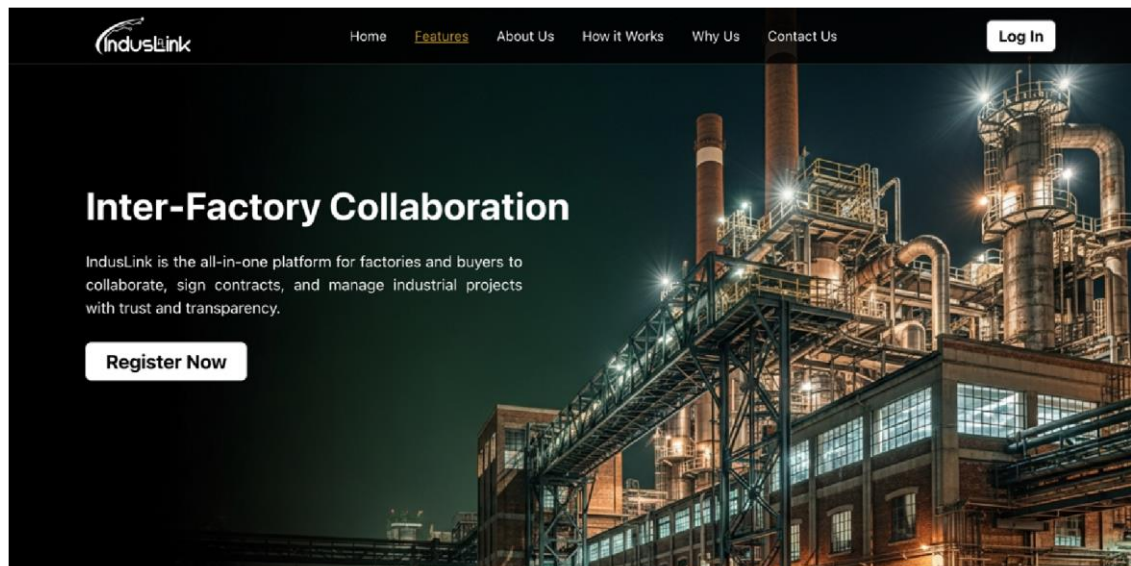
5.3 User Interface

This is part of system which facilitates the user to access features of the system visually and interactively. Basically in IndusLink we use the Sequential Approach to build our system because every module depends on its previous one. The interface is divided into two main sections one is buyer side and second one is seller side. Both sections have their own features. Firstly, user will register themselves. Then login page for the user authentication. In dashboard user can view their own activities and notifications. In contract module buyer can create contract and seller can manage it. Orders module help users to manage their orders and to send and to receive the orders. Chat system help buyer and seller to communicate in real time. In every component forms, table and buttons are used to help users interact effortlessly.

Programming languages: JavaScript (with JSX syntax)

Framework: Next.js 16

Libraries: Shadcn/UI and Radix UI, Iconify React, Lucide React, React Icon, Tailwind CSS, Recharts, React Toastify, Emoji Picker React



Our Features



Seller & Buyer Profiles

Engage more customers by showing their abilities & credibility.



Digital Contract System

Enhance transparency by eliminating physical visits & traditional paper-based agreements.

logo

AI-Based Service Matchmaking

Saves time by suggesting the best fit service providers instead of typical searching.



Ordering System

Complete history of all orders against single contract enhance visibility.



Integrated Chat System

Communicate instantly with your business partners that minimize verbal communication.



Role-Based Interface

Access only relevant features and data which increase flexibility.

About Us

Our Story

Born from real industry challenges, IndusLink bridges the communication gap between factories. We simplify how businesses find trusted partners through physical visits, negotiate, and manage contracts, all in one unified platform to connect, collaborate, and grow.



About IndusLink

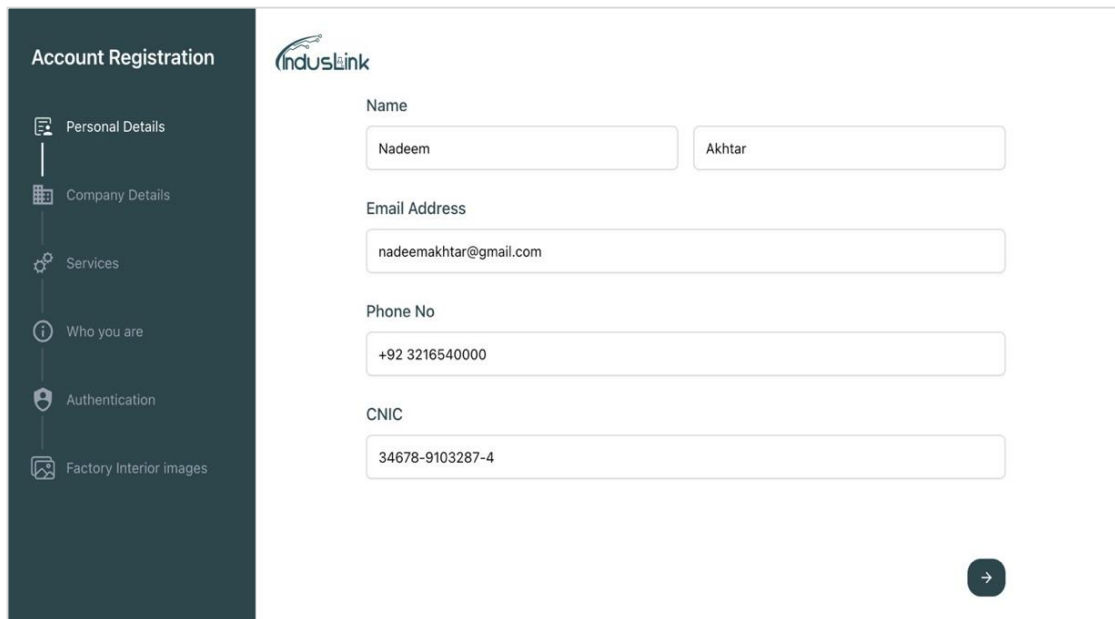
IndusLink was founded with a clear vision to transform industrial collaboration making it simpler, smarter, and more transparent. Our mission is to help businesses save time, cut costs, and build trusted partnerships through a seamless, secure, and scalable digital platform.

Figure 5.1 Marketing Page



Figure 5.2 Marketing Page

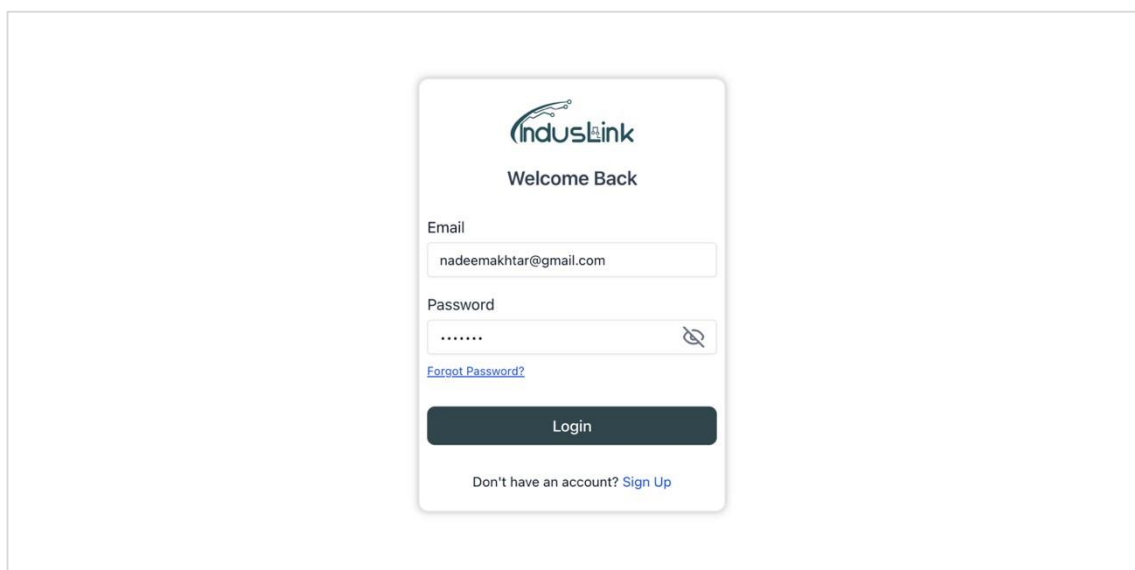
Figure 5.1 and 5.2 show the marketing page of IndusLink to attract the companies.



The image shows the 'Account Registration' page for IndusLink. On the left is a dark sidebar with a vertical list of steps: 'Personal Details' (selected), 'Company Details', 'Services', 'Who you are', 'Authentication', and 'Factory Interior images'. The main content area has the IndusLink logo at the top. Below it are input fields for 'Name' (split into 'Nadeem' and 'Akhtar'), 'Email Address' (nadeemakhtar@gmail.com), 'Phone No' (+92 3216540000), and 'CNIC' (34678-9103287-4). A dark 'Next' button with a right arrow is at the bottom right.

Figure 5.3 Registration Page

In Figure 5.3, Companies has to register their companies by following all steps.



The image shows the 'Login' page for IndusLink. It features the IndusLink logo and the text 'Welcome Back'. There are input fields for 'Email' (nadeemakhtar@gmail.com) and 'Password' (masked with dots). A 'Forgot Password?' link is below the password field. A dark 'Login' button is centered below the fields. At the bottom, it says 'Don't have an account? Sign Up' with a link.

Figure 5.4 Login Page

In Figure 5.4, Companies has to login to use the system and access their data.

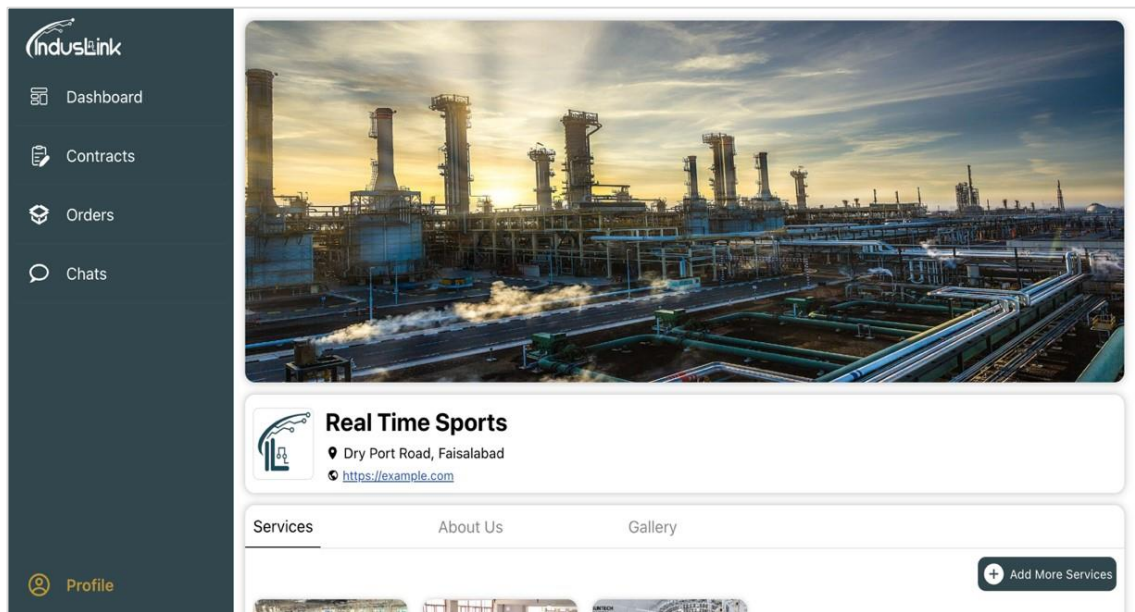


Figure 5.5 Profile Interface

In Figure 5.5, buyer can edit and update his profile.

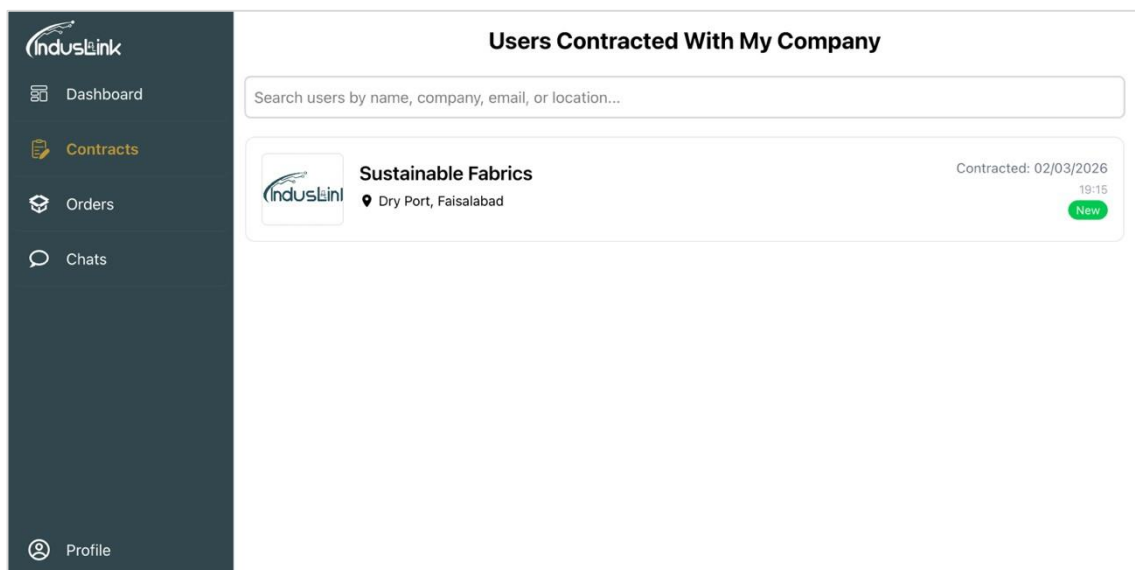


Figure 5.6 Contracted Companies

In Figure 5.6, buyer can see the companies with which he wants to start contract.

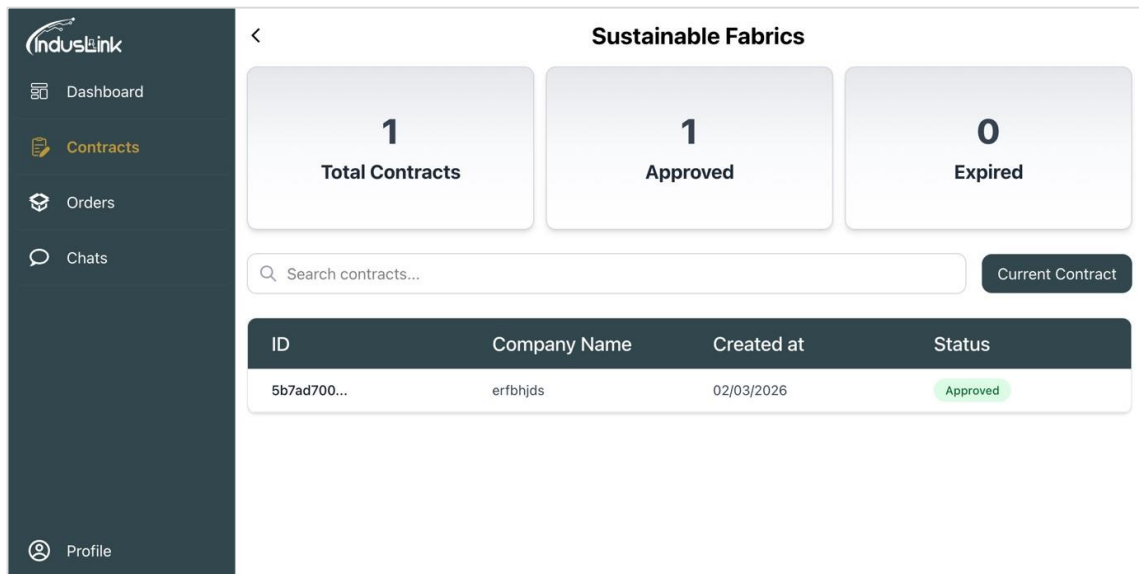


Figure 5.7 Selected Company

In Figure 5.7, buyer and seller can view their contract history with each company.

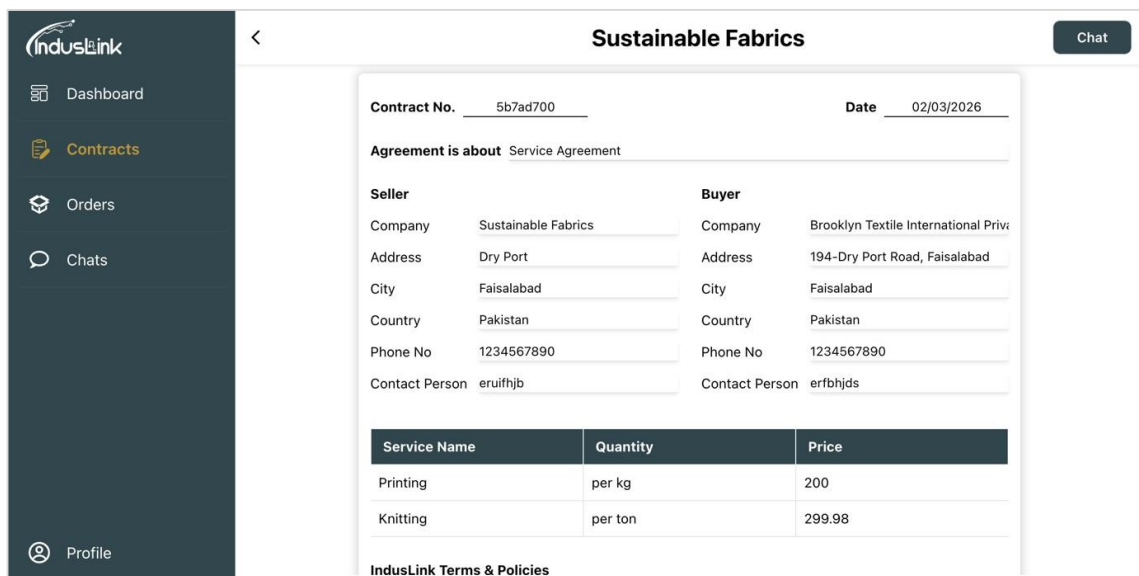


Figure 5.8 New Contract

In Figure 5.8, buyer can add new contract and send it to the seller. Also they can use integrated chat system in contract. Each contract has their own chat.

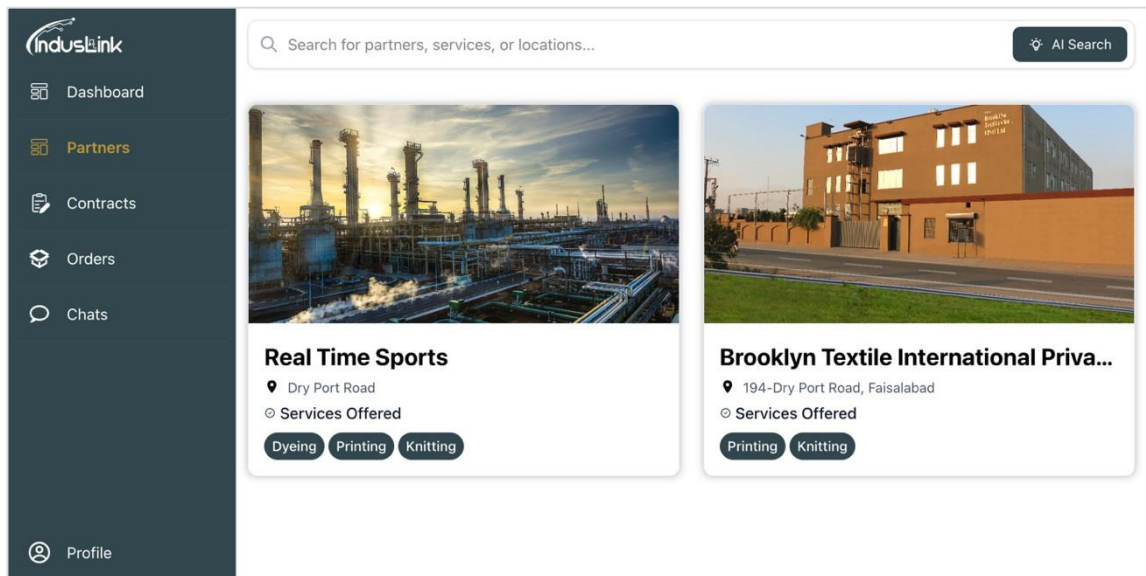


Figure 5.9 Partners Interface

In Figure 5.9 buyer can find the best suited seller by exploring the their profiles.

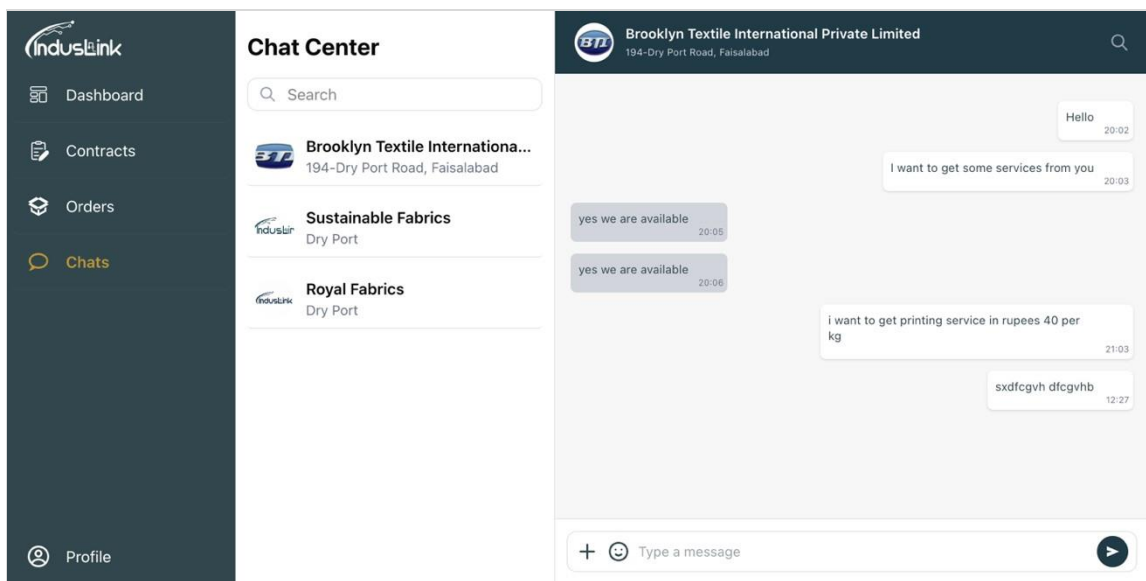


Figure 5.10 Chat Interface

In Figure 5.10, seller and buyer can send and receive messages to each other by using IndusLink chat system.

Order ID	Date	Status	Amount	View	Invoice	Actions
88707128...	17/05/2026	Delivered	Rs. 6,800	View	View Invoice	Mark as Received
1e21ed91...	17/05/2026	Approved	Rs. 2,000	View	View Invoice	Waiting for Delivery
aa13ad23...	17/05/2026	Approved	Rs. 6,000	View	View Invoice	Waiting for Delivery
fbdb791d...	17/05/2026	Pending	Rs. 40,600	View	Not available	Waiting for Delivery

Figure 5.11 Orders List

In Figure 5.11, complete orders history is showed to buyer on buyerside and to seller on sellerside according to selected company.

Description	Unit Price	Quantity	Amount
Select a service	Rs. 0.00	0	Rs. 0.00

Items Total: Rs. 0.00

Total: Rs. 0.00

Figure 5.12 Order Creation

In Figure 5.12, buyer create the order and send it to seller. He can also add more items by clicking on Add New Item button.

IndusLink

Dashboard

Contracts

Orders

Chats

Profile

Generate Invoice

Brooklyn Textile International Private ...

194-Dry Port Road, Faisalabad, Faisalabad

Phone: +923216540000

Email: Not provided

INVOICE

#INV-fbdb791d-2026

Date: May 17, 2026

Bill To:

Company

Sustainable Fabrics

Phone

+92 3216540000

Address

Dry Port, Faisalabad

Email

toobaattaria12@gmail.com

Description	Unit Price	Quantity	Amount
Printing	200	203	Rs. 40600.00

Notes / Instructions

No notes provided for this order

Subtotal:

Rs. 40600.00

Discount(%):

0

- Rs. 0.00

Tax rate(%):

0

Rs. 0.00

Figure 5.13 Invoice Generation

Figure 5.13 display the invoice UI. When seller receives the orders then system generate the invoice automatically and show to buyer.

CHAPTER 6: SYSTEM TESTING

6.1 Manual Testing

In Manual Testing, testers performed testing of the system manually interacted with the interface of system. The main goal is to evaluate the usability, reliability and functionality of the IndusLink. This all process is manual without using any automated tools.

Manual testing was performed to determine the functionalities of each module of IndusLink that work correctly under different circumstances, IndusLink should behave correctly in both valid and invalid inputs of the user and error detection in system. Manual testing was performed on modules of IndusLink such as registration, login, profile management, contract creation, contract approval, and chat system.

6.1.1 System Testing

System testing was performed to evaluate every module of IndusLink together as whole system. The main goal is to focus on validating user workflow.

User Registration

User open the IndusLink marketing page and click on Register now button then user enter all the inputs and click on submit.

Scenarios:

1. User clicks on submit button while input fields are empty.

Expected Result: The system should display an error message.

Actual Result: The system successfully showed an error message.

2. The user fill all inputs fields and click on submit button.

Expected Result: The system should register the user.

Actual Result: The system successfully registered user.

3. If the email already exists.

Expected Result: System should show an error message that user already has registered.

Actual Result: The system successfully showed an error message.

User Login

After login the user login to his account by entering the registered email & password.

Expected Result: The dashboard appears Actual

Result: Dashboard appears successfully.

Scenarios:

1. If user enter the wrong password or email.

Expected Result: The system should show an error message.

Actual Result: The system showed an error message successfully.

2. If user forgot the password.

User clicks on forgot password, enter email and receive an otp on his email then user enter an otp.

Expected Result: The user's password should be changed. Actual

Result: The user's password changed successfully.

Profile Management

When user clicks on profile button from side panel the profile page of the user appears.

Scenarios:

1. Data fetching in profile after registration

Expected Result: User Data should be fetched in profile after registration.

Actual Result: Data fetched successfully.

2. User wants to edit the profile such as profile image, services, gallery images, about.

Expected Result: Profile information should be updated.

Actual Result: Profile updated successfully.

3. User wants to delete services, images

Expected Result: System asks for confirmation should deleted the data.

Actual Result: Data deleted successfully.

Real Time Chatting

User clicks on the chat button from side panel and chat page appears.

Scenarios:

- 1. User (seller, buyer) send message to another user (seller, buyer).**
Expected Result: Messages should be delivered.
Actual Result: Messages successfully exchanged.
- 2. Buyer see the list of sellers and seller should see the list of buyer in chatting system.**
Expected Result: Both buyer and seller should see each and another lists of contact in the chat module.
Actual Result: List shown successfully.

Contract Management

Buyer visits the seller profile and clicks on the start contract button. The contract page appears.

Scenarios:

- 1. Buyer clicks on start contract button and fill all the fields in the contract.**
Expected Result: The contract should be shown to seller.
Actual Result: The contract is successfully shown to seller.
- 2. Seller fill all the field from his side and click on done button.** Expected Result: System should update the contract to buyer Actual Result: Contract updated successfully to buyer.
- 3. If Buyers wants to edit anything clicks on edit and save button.** Expected Result: System should update the contract to seller Actual Result: Contract updated successfully to seller.
- 4. Buyer clicks on approve button and enter password.**
Expected Result: System should verify the password and send opt to the email of buyer.

Actual Result: System verified the password and sent opt successfully and approved the contract from buyer side.
- 5. Contract appears to seller and seller approve it.**
Expected Result: System should verify the password and send opt to the email of seller.

Actual Result: System verified the password and sent opt successfully and approved the contract from seller side.

6. Once contract approved either seller or buyer cannot alter

Expected Result: The system should not allow to alter the contract

Actual Result: The system successfully not allowed the users to alter the contract.

7. The user entered the wrong password or otp.

Expected Result: System should not verify the user.

Actual Result: The system successfully not verified the user.

Order Management

Order management module provides secure order creation and invoice generation against the specific contracts.

Scenarios

1. Buyer can create order from an approved contract

Expected Result: Order should be created and should appear on seller dashboard.

Actual Result: Order created and displayed successfully.

2. Seller can accept order

Expected Result: Seller can accept an order.

Actual Result: Seller successfully accepted an order.

3. Buyer can view invoice

Expected Result: Seller can generate an invoice and buyer can view the generated invoice from seller.

Actual Result: The invoice successfully visible to seller.

4. Seller can update order status (production, dispatch) Expected Result: Seller can update status.

Actual Result: Seller successfully updated order status.

5. Buyer will mark order as delivered

Expected Result: Buyer can mark order as delivered and order status will mark as completed from both side.

Actual Result: Successfully order completed.

AI-based Service Match-Making

AI module suggests factories according to the given prompt also provide the reason why I suggest you.

Scenarios

1. **Buyer click on AI search button and give prompt** Expected Result: Buyer should give prompt successfully.
Actual Result: Successfully gave prompt.
2. **AI Process the prompt and show filtered factories**
Expected Result: AI should process the prompt and show filtered factories.
Actual Result: Successfully processed the prompt and showed factories by AI.
3. **AI provides reason for recommended factories**
Expected Result: AI should provide the reason for recommended factories.
Actual Result: Successfully provided the reason.

6.1.2 Unit Testing

Unit testing verifies the functionalities of individual modules of the system independently. We tested each module of IndusLink separately to detect and correct any fault early in the system.

Unit Testing Methodology

- **Input Validation Testing:** It validates the user inputs.
- **Boundary Testing:** It validates the edge cases such as invalid and empty fields.
- **Positive Testing:** User provides the valid inputs to ensure the system behaves as expected.
- **Negative Testing:** User provides the invalid inputs to ensure system handles errors.

Unit Test Cases

Registration

1. **Missing fields in Registration**
Expected Result: The system should show an error message.
Actual Result: The system successfully showed an error message.
2. **The user fill all inputs fields in Registration.**
Expected Result: The system should register the user.
Actual Result: The system successfully registered the user.

Login

1. Login with valid credentials

If user enter the valid password or email.

Expected Result: The system should login the user.

Actual Result: The system login the user successfully.

2. Login with invalid credentials

If user enter the wrong password or email.

Expected Result: The system should show an error message.

Actual Result: The system showed an error message successfully.

Contract Management

1. Valid password in contract verification

The user entered the valid password or otp.

Expected Result: The system should verify the user.

Actual Result: The system successfully verified the user.

2. Invalid password in contract verification

The user entered the wrong password or otp.

Expected Result: The system should not verify the user.

Actual Result: The system not verified the user successfully.

Order Management

1. Buyer can create order from an approved contract

Expected Result: Order should be created and should appear on seller dashboard.

Actual Result: Order created and displayed successfully.

2. Buyer submit empty fields during order creation Expected

Result: System should not submit an order.

Actual Result: System successfully not submitted an order.

3. Seller can accept order

Expected Result: Seller can accept an order.

Actual Result: Seller successfully accepted an order.

AI-based Service Match-Making

1. AI Process the prompt and show filtered factories

Expected Result: AI should process the prompt and show filtered factories.

Actual Result: Successfully processed the prompt and showed factories by AI.

2. AI provides reason for recommended factories

Expected Result: AI should provide the reason for recommended factories.

Actual Result: Successfully provided the reason.

6.1.3 Functional Testing

Functional testing verify that every feature of the system behaves according to the defined requirements. The main focus of functional requirement is to verify that the functions of all available features should behave correctly in valid or invalid inputs of the users. For the IndusLink, we tested each function by provided the predefined test cases to verify that actual result should matched with the expected results.

Functional Test Cases

1. Register user with valid input

Expected Result: The system should register the user.

Actual Result: The system successfully registered the user.

2. Register user with incomplete data

Expected Result: The system should show an error message.

Actual Result: The system successfully showed an error message.

3. Login with valid credentials

Expected Result: The system should login the user.

Actual Result: The system login the user successfully.

4. Login with invalid credential

Expected Result: The system should show an error message.

Actual Result: The system showed an error message successfully.

5. Update or edit the profile

Expected Result: Profile information should be updated.

Actual Result: Profile updated successfully.

6. User (seller, buyer) send message to another user (seller, buyer) Expected

Result: Messages should be delivered.

Actual Result: Messages successfully exchanged.

7. Buyer see the list of sellers and seller should see the list of buyers in chatting

Expected Result: Both buyer and seller should see each and another lists of contact in the chat module.

Actual Result: List shown successfully.

8. Contract verification with valid password The user entered the valid password or otp.

Expected Result: The system should verify the user.

Actual Result: The system successfully verified the user.

9. Contract verification with invalid password The user entered the wrong password or otp.

Expected Result: The should not verify the user.

Actual Result: The system not verified the user successfully.

10. Data Updating after the seller and buyer interaction with the contract.

Expected Result: Contract should be updated. Actual

Result: Contract updated successfully.

11. Seller can update order status (production, dispatch) Expected Result: Seller can update status.

Actual Result: Seller successfully updated order status.

12. Buyer will mark order as delivered

Expected Result: Buyer can mark order as delivered and order status will mark as completed from both side.

Actual Result: Successfully order completed.

13. AI Process the prompt and show filtered factories

Expected Result: AI should process the prompt and show filtered factories.

Actual Result: Successfully processed the prompt and showed factories by AI.

Challenges faced in System during Functional Testing:

During testing there was slight delay in the chat messages delivery.

Results

All features of IndusLink worked properly according to the defined requirements. There were slight delays in chat messages that was addressed by optimizing backend processing.

6.1.4 Integration Testing

Integration testing verify that different modules of the system work correctly when combined together. The main focus is to ensure that data flows correctly between the different modules and interaction between them remains smooth. For IndusLink we tested that all modules of the system communicate properly and as per expectation and data flow between them correctly.

Integration Test Scenarios

Registration and login

1. **User registered and log in to the account**

Expected Result: User should log in after registration.

Actual Result: User log into the system successfully after registration.

2. **User log into Account and Dashboard appears** Expected Result:

System should display dashboard.

Actual Result: System displayed dashboard successfully.

Registration and Data fetching in Profile

1. **Data fetching in profile after registration**

Expected Result: User Data should be fetched in profile after registration.

Actual Result: Data fetched successfully. **login**

and Profile Management

1. **User logs in and can update profile**

Expected Result: User should log in and then can edit or update profile.

Actual Result: User log into the system and updated profile successfully.

Profile Management and Partners

1. **The user adds details in the profile and can search the partner accordingly.**

Expected Result: User should find partner according to profile description.

Actual Result: User found partner successfully.

Partner and Contract

Expected Result: Buyer can create contract after finding partner by clicking on start contract.

Actual Result: Buyer created contract successfully.

Integrated Chat System

Expected Result: Seller and buyer can chat within the contract module.

Actual Result: Seller and buyer did chat successfully.

Contract and Order

Expected Result: Only approved contract can visible in order module.

Actual Result: Successfully visible in order module.

Order created and seller side

Expected Result: Buyer created an order and appears to seller.

Actual Result: Successfully appeared to seller.

Seller acceptance and send invoice to buyer

Expected Result: Seller accept an order and generate invoice and send it to buyer.

Actual Result: Buyer received an invoice successfully.

Seller update order status and visible to buyer side

Expected Result: Seller can update order status and buyer sees the latest order status.

Actual Result: Buyer saw latest order status successfully.

Partner and AI-based Service Match-Making

Expected Result: In partner module buyer can search for AI-based partner.

Actual Result: Successfully buyer searched AI-based partner.

Integrational Issues

During the early phases of testing, the chat messages required page reload to appear and profile information was not appearing immediately after the updating the profile.

We fixed these issues by optimization of data handling and proper data synchronization from backend.

6.2 Automated Testing

Automated testing was performed to verify the reliability of system and to ensure efficiency of testing process. Manual testing is the time taking process where the testers

create test cases manually and executed by the human but in automated testing, we use the scripts and automated tools that run the system automatically to ensure the behavior of system is correctly working.

Automated testing complements manual testing by its process such as manual testing is used to ensure the usability and good user experience but it is time taking process while, automated testing is used for executing repetitive tasks efficiently and consistently also it reduces the chances of human errors. We use selenium for the process of automated testing.

Automated Test Scripts

Login

```
from selenium import webdriver

from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time

openDriver = webdriver.Chrome()
wait = WebDriverWait(openDriver, 20)

openDriver.get("http://localhost:3000/login")
userEmail =
wait.until(EC.presence_of_element_located((By.NAME, "email")))
userEmail.send_keys("induslink@gmail.com")
userPassword =
openDriver.find_element(By.NAME, "password")
userPassword.send_keys("induslink@IN")
userPassword.send_keys(Keys.RETURN)

time.sleep(10)

openDriver.quit()
```

Profile

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time
```

```

openDriver = webdriver.Chrome() wait = WebDriverWait(openDriver, 20)

openDriver.get("http://localhost:3000/login") userEmail =

wait.until(EC.presence_of_element_located((By.NAME, "email")))

userEmail.send_keys("induslink@gmail.com") userPassword =

openDriver.find_element(By.NAME, "password")

userPassword.send_keys("induslink@IN")

userPassword.send_keys(Keys.RETURN)

wait.until(EC.url_contains("/sellerside"))

profile          =          wait.until(EC.presence_of_element_located((By.XPATH,
"//a[@href='/sellerside/profile']")))

openDriver.execute_script("arguments[0].scrollIntoView();",
profile) openDriver.execute_script("arguments[0].click();", profile)

wait.until(EC.url_contains("/sellerside/profile")) time.sleep(20)

openDriver.quit()

```

6.2.1 Tool Used

We used Selenium to performed the automated testing of our system.

Why Use? (benefits)

Selenium was chosen for automated testing because it supports web application testing and IndusLink is web-based application, selenium is compatible with multiple browsers, and it also automates the repetitive tasks that reduces the time and effort of the tester.

6.3 Results & Discussion

The focus of testing was to ensure the reliability of the system in different test cases of user interaction. Our objective was to meet the system requirements, to ensure that all the functions of each module should work correctly and to ensure the smooth collaboration between the buyer and seller.

Both Manual and automated testing was conducted during testing for smooth user interaction, workflow and automated repetitive operations work efficiently for the

verification of the system. We covered all major modules of IndusLink during testing such Registration, login. Profile management, contract, and chat system.

Testing Results

We executed all pre-defined test cases, automated testing and result confirmed all functionalities of each module working as expected and system is fully capable of user interaction such as user registration, login, profile management, contract, and chat system.

Test Cases Result Table

Table 6.1 Test Cases Result Table

Testing Type	Total Test Cases	Passed	Failed
System Testing	25	23	2
Unit Testing	11	10	1
Functional Testing	13	11	2
Integration Testing	11	10	1

Performance Metric

Here are some metrics that are used to identify the stability of the system.

- **Test cases:** Around 90% of executed test cases passed successfully with no error.
- **System Stability:** These passed executed test cases has approved that system is stable for the interaction of real user.
- **Problem Detection:** The minor problem was detected in the early stages of testing that was solved successfully.
- **Result Accuracy:** Our system approximately, passed all test cases by matching the actual result with the expected result.

Effectiveness of Testing Strategies

We combined both manual and automatic testing, this strategy has ensured smooth user interaction and proper backend operations such as proper data synchronization.

Moreover, these testing helped to identify errors in the early stages of testing and to verify the system functionalities working correctly.

Improvements

During the testing process several issues were detected such as no proper data synchronization, and delay in chat messages delivery. We fixed these issues by the proper implementation of backend data operations. So, after the improvements in our system, we have ensured that our platform is reliable, stable and easy to use for the real time users.

CHAPTER 7: CONCLUSION

Problem Faced and Lessons Learned

AI service Matchmaking

- **Problem Faced:** We faced issues in prompt designing & response when we use LLM (Gemini 1.5 Flash) for AI service matchmaking. Model sometimes didn't follow the proper output structure & sometimes gave generic data.
- **Lessons Learned:** we learned efficient prompting and context handling which is very important for effective results. We also gained the practical experience in response handling.

Supabase – Database Backend

- **Problem Faced:** In supabase we faced challenges in database tables, relationships and to import data from CSV and we also faced issues to add dummy data in our system.
- **Leason Learned:** We learned and gained experience to design and manage Relational database. We also learned to run SQL quires to insert dummy data in case if the CSV fails to import.

Conclusion

The proposed IndusLink platform aims to transform the traditional processes of textile industry into digitalized inter-factory collaboration by addressing the challenges such as find specific service provider, digital contract approval and communication. This platform eliminates the need for time-consuming physical visits and ensures transparency by integrating detailed business profiles, role-based access, intelligent matchmaking, digital contracts approval, secure ordering and messaging system. Our IndusLink platform will revolutionize the traditional textile sector of Faisalabad into modern digital industry by enhancing the efficiency, transparency and trust.

Limitations of the Project

- Our System is designed only for textile sectors, expansion to other industries will require additional customization.

- Payment module is not integrated in our system. Users follow the manual payment methods for transactions.

Future Work

Currently our system is designed only for textile sector but in future we will expand our system to support multiple sectors and industries so that our system can be more useful and flexible.

In future we will integrate the secure and efficient payment module so that users can make online payments that will make transactions faster and more reliable. This will make our system more efficient and user-friendly.

References

- FCCI** (2024) *FCCI*. Available at: <https://fcci.com.pk/> (Accessed: 22 August 2025).
- Alibaba** (2025) *Alibaba*. Available at: <https://www.alibaba.com/> (Accessed: 11 October 2025).
- Fibre2Fashion** (2019) *B2B Marketplace, B2B Business Solutions, Business Directory*. Fibre2fashion.com. Available at: <https://www.fibre2fashion.com/> (Accessed: 11 October 2025).
- TradeKey** (2025) *Pakistan B2B Marketplace*. Pakistan.tradekey.com. Available at: <https://pakistan.tradekey.com/> (Accessed: 11 October 2025).
- Zahra, M., Naqvi, S.A.A., Hussain, B. and Magazzino, C.** (2025) Digitalizing sustainability in Pakistan's textile sector: An investigation of lean digital transformation adoption. *International Journal of Engineering Business Management*. Available at: <https://doi.org/10.1177/18479790251369174> (Accessed: 3 November 2025).

